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Quantifying the weight of evidence from a forensic fingerprint comparison: a new paradigm

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Summary. The fingerprint has, with considerable justification, come to be regarded as the acme of forensic identification. Over the last century, millions of cases have been resolved world wide because of marks left at crime scenes. The comparison methodology has not evolved greatly during its history and it is universal practice to present fingerprint evidence to a court as a categoric opinion of identification or exclusion, or to classify the evidence as inconclusive and not to report it. There has been a growing movement to supplement the fingerprint examination process by one that has a statistical model, supported by appropriate databases for calculating numerical measures of weight of evidence. The movement calls for the establishment of a logical framework for informing conclusions, based on explicit assumptions and data and open to revision and improvement. The aim is to enable the numerical evaluation of evidence that would currently be reported as a categorical identification and also of evidence that would currently be classified as inconclusive. The paper presents the results of a project carried out by the Forensic Science Service that aims to attain this goal. After a historical review, we describe a formal model for assigning numerical values to configurations of *minutiae* in fingerprints. We describe how the parameters of the model have been optimized to take account of interoperator variability and distortion of the finger pad, and we present the results of a substantial validation experiment that was based on searches that have been carried out on the US national fingerprint database of approximately 600 million fingerprints.

Keywords: Criminal justice; Fingerprints; Forensic science; Likelihood ratio; Weight of evidence

1. Introduction

The wealth of information that is exhibited in the friction ridge pattern of an individual fingerprint is universally acknowledged and is extensively used for the detection of offenders. In 2002, more than 270000 requests for fingerprint examinations were received by publicly funded forensic laboratories and bureaux in the USA (Peterson and Hickman, 2005). In the same year, fingerprint evidence was recovered from approximately 330000 crime scenes in the UK, leading to the detection of approximately 34000 suspects (UK Home Office, 2005).

Given a mark from a scene of a crime, it is helpful to distinguish between two processes: one of retrieval of a candidate fingerprint from a database for investigation purposes; and one of comparison between the print from a known individual and the crime mark to evaluate evidence

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for potential court purposes. It is the latter process that is considered in the present paper. It comprises several stages. An examiner first studies the crime mark carefully and then looks at the print from the known individual to consider the overall quality of the mark and the quantity of both similarities and differences with a view to forming an opinion with regard to the identity of the source. Usually, the opinion (and sometimes the process) is verified by one or more additional examiners.

The current culture within fingerprint communities world wide demands that fingerprint identifications should be presented at court as categorical opinions. A fingerprint examiner will tell the court that, on the basis of a careful and detailed comparison, he or she is *absolutely certain* that a particular mark was made by the person who provided the control print to the exclusion of all other individuals on Earth (Saks and Koehler, 2008; Cole, 2009). The achievement of that state of certainty, as Saks, Koehler and Cole allude, is a question of epistemology: it cannot be rationalized scientifically, except in trivial situations, and leads to an overstatement of the evidence when presented in court. The last decade has seen a growing body of literature that questions this paradigm.

Positive opinions aside, a large proportion of comparisons will lead to negative opinions, which tend to be non-controversial. There is a third class of comparisons: those where the examiner considers, for various reasons, that the evidence is of insufficient quality, or comprises too little detail for an opinion to be formed. This means that there is a class of cases where evidence which might be of corroborative value is not taken further because the examiner considers it to be insufficient for an opinion of complete certainty. This reasoning has obvious benefits in relation to quality assurance, casework turnaround time and costs; nevertheless, it means that evidence of potential value to a court is lost.

We do not challenge the overall reliability of the conclusions of the immense majority of fingerprint comparisons performed over the past century. Nor do we intend to minimize their contribution to the criminal justice system. However, we agree with views expressed in the literature (Zabell, 2005; Office of the US Inspector General, 2006; Saks and Koehler, 2005, 2008; National Research Council of the National Academies, 2009) that the evaluation of the weight of evidence associated with any particular fingerprint comparison lacks both a scientific foundation and transparency.

When it comes to deoxyribonucleic acid (DNA) profiling—arguably the most important advance in forensic science since fingerprints—the judgement in R *versus* Doheny and Adams (see, for example, Evett *et al.* (2000)) directs that weight of evidence in relation to the issue of identity of source must be imparted by means of a statistic. This creates a paradox where courts accept that fingerprint examiners form and express categorical opinions, whereas DNA specialists must indicate weight of evidence by means of a statistic that is computed from genetic principles and an appropriate reference database.

In this paper, we describe research that has been carried out by the Forensic Science Service to develop a method for assigning a quantitative measure of weight of evidence to a fingerprint comparison. We see this as creating the kind of scientific foundation that will

(a) inform and support conclusions,
(b) contribute to quality management functions, such as training and standardization between individuals and organizations, and
(c) provide measures of weight of evidence in cases that, at present, would be considered to contain insufficient detail for an opinion.

The next section presents a brief review of the history of forensic fingerprint comparison. Section 3 explains a basic system of notation for a generalized evidential transfer problem.

Section 4 describes the method that has been developed for extracting quantitative data from a fingerprint. In Section 5, we outline the databases that have formed the basis of the present study. Section 6 is a brief description of the normalization of the metric that is used to compare fingerprint data. Section 7 shows how a distortion model is used to generate simulated crime marks. In Section 8, we show how the generalized solution to the transfer problem, which is described in Section 3, can be applied to the fingerprint comparison problem. In Section 9, we describe the work that has been carried out to validate the method and give results from tests against the largest fingerprint database in the world. In Section 10, we discuss important inferential issues and we explore the scope for future work. Finally, in Section 11, we discuss the immediate implications of our results for the fingerprint community and the potential for a paradigm shift in the world of fingerprint identification.

2. Historical review

The use of fingerprints for personal identification was first established in ancient China and India. From 1858, Sir William Herschel promoted the use of fingerprints to authenticate contracts, deeds, payments and warrants (Herschel, 1916). In 1863, a Frenchman, Paul-Jean Coulier, proposed a technique to develop latent fingerprints and suggested their use for the identification of criminals (Margot and Quinche, 2010). In parallel to Herschel, Dr Henry Faulds realized the usefulness of fingerprints and proposed that they may lead to the scientific identification of criminals (Faulds, 1880).

Sir Francis Galton is usually credited with the first extensive research published on fingerprints (Galton, 1892). Galton collected more than 8000 sets of 10 fingerprints and showed that pattern and friction ridges are persistent over time. In an attempt to demonstrate the specificity of friction ridge detail, he proposed the first statistical model for fingerprints (Stoney, 2001).

Since then, a fragmentary finger impression recovered from the scene of a crime is known, to all British fingerprint examiners, as a *mark*; a finger impression taken from a suspect under controlled conditions is known as a *print*. Note that various regions of the body are covered with a characteristic friction ridge surface—fingers, palms, toes and the soles of the feet. However, most scene-of-crime marks are made with the first joint of the finger and it is such marks that we consider in this paper. The extension of the theory to marks from other regions is not trivial, but it is entirely feasible.

The pattern of the ridges (Fig. 1) that is observed in a print can be classified by nomenclature that was first adopted by Sir Edward Henry (Henry, 1900) and such classification systems formed the basis of manual retrieval from databases of the prints of individuals who had come to notice from previous crimes. We are not primarily concerned with such patterns in this paper; nor do we consider the retrieval of prints from databases. We are concerned with a finer level of detail that is used by examiners in arriving at opinions that may be used at court. An individual ridge in a print is subject to two perturbations: it might end; or it might divide into two (exceptionally into three). Such points of detail are called *minutiae*, or points, and it is the comparison of the relative positions of a set of pairs of *minutiae* between a mark and a print that informs examiners' opinions (Fig. 1).

Between 1911 and 1914, Edmund Locard established the first set of rules for identifying a mark with a print using these features (Locard, 1914). Locard's rules suggested that, if there were 12 points of agreement and no differences, this established identity beyond doubt. He also implied that if there were fewer points of agreement then a probabilistic opinion might be justified.

The qualitative aspects of Locard's rules were largely ignored and a 12 points standard was adopted in most countries as a criterion for stating a categorical opinion of identity of source for



Fig. 1. Main patterns of ridge flow (◇, core (centre) of the patterns; Δ, delta (point of convergence of the ridge flow) of the patterns): (a) whorl; (b) arch; (c) loop

a mark and print—usually presented as the *individualization of the source of the mark to the exclusion of all other sources* (Kirk, 1963), by the fingerprint community (Saks and Koehler, 2008).

The situation in the USA developed rather differently. Although examiners were influenced by Locard's rules, they gradually moved away from the notion of a standard based on counting points. A review that was commissioned in 1973 by the International Association for Identification concluded that

> 'no valid basis exists for requiring that a predetermined minimum number of friction ridge characteristics must be present in two impressions in order to establish identity'.

This resolution was reconfirmed by the Association in 2009. US examiners did not adopt Locard's suggest for probabilistic opinions. A resolution that was passed by the Association in 1979 (and finally rescinded in 2010) prohibited its members from expressing any probabilistic opinion in relation to a fingerprint comparison, on pain of expulsion from the Association.

Since 1926, UK fingerprint bureaux had worked to standards that required 16 corresponding points to be marked up in mark and print. In 1988, the Home Office commissioned a study (Evett and Williams, 1996) to investigate whether or not there would be advantages in moving from the 16 points standard to some other numerical standard. A collaborative study involving about 130 British fingerprint examiners, which was carried out as part of the review, revealed a large variation between examiners in the number of points called in a given comparison. There was also substantial variation between examiners in their judgements about whether or not a given comparison could be classified as an identification. The conclusion of the review was that a national system of quality management was needed in the UK. Although Evett and Williams made no specific recommendation in relation to the 16 points standard, it was officially abandoned in 2001, following a directive from the Association of Chief Police Officers.

History over the past century shows that fingerprint practitioners have relied on very limited systematic research to inform their evaluation of the weight of fingerprint evidence and to form their conclusions. The existing approaches are heuristic and the reasoning that leads

to an identification is essentially a psychological one that cannot be rationalized and rendered explicit. In recent years, fingerprint evidence has been challenged in courts (mostly in the USA—Cole (2004)), and in the legal and scientific literature (Zabell, 2005; Saks and Koehler, 2005, 2008; National Research Council of the National Academies, 2009; Cole, 2005, 2009). General criticisms of the discipline follow three broad lines of argument.

(a) The selection process of fingerprint features, and the subsequent use of those features to form opinions of identity about the source of fingerprints, lacks reproducibility and is difficult to peer-review.
(b) Claims for the logic of a categorical opinion of identity tend to invoke the argument that all fingerprints are unique. A complete print taken under controlled conditions from an individual undoubtedly bears a large amount of information and there is a strong argument for believing that this is sufficient to distinguish it from all other control prints that ever existed. However, the weight of evidence to be assigned to a crime comparison is dominated by the information content of the mark but marks recovered from crime scenes are often limited in extent, of poor quality, subject to distortion and also to interference from the substrate. The claim of the uniqueness of fingerprints becomes then irrelevant to fingerprint examination and even more so to the forming of the conclusions. It follows from this argument that fingerprint examiners overstate the weight of evidence when reporting their findings.
(c) The reliability of fingerprint identification has not been measured. Members of the fingerprint community have claimed for many years that the error rate of fingerprint examination was zero (Cole, 2005). However, well-publicized errors or disputed identifications (Office of the US Inspector General, 2006; Cole, 2005), have recently been brought to light. Some research has been initiated on this topic (Langenburg, 2009; Wertheim *et al.*, 2006) and the early results suggest that the error rate is low. Nevertheless, there is little hard data to enable a court to assess the reliability of an opinion in an individual case.

There is no doubt that the fingerprint professions in both the UK and the USA have taken major strides in training, competency assessment and standardization. The central issues of the evaluation of the weight of evidence of a fingerprint comparison and, hence, of the forming and reporting of balanced conclusions remain. Much good work has been done to develop statistical models and the research that we have carried out builds on progress made by others. An excellent review of early work was given by Stoney (2001). Weaknesses of early models included reliance on untested assumptions of independence, failure to take account of the spatial relationships of fingerprint features, failure to take account of distortion and failure to take account of variability between examiners. More recent models by Champod (1995), Neumann *et al.* (2006, 2007) and Egli *et al.* (2006) have been directed at these factors. The present paper describes an evolution of the model that was proposed by Neumann *et al.* (2007).

3. Notation and basic theory

The essence of the problem that we consider is similar to that addressed by Lindley (1977) and we first distil that essence by describing a generalization of a very common forensic situation. Imagine that there is an item of evidence, a trace, the origin of which is a matter of dispute between two parties: prosecution and defence. There is another item of evidence, a control, which is of undisputed origin. To provide evidence that might contribute to the resolution of the dispute, observations are made of a series of properties of the trace. These observations may take various forms (numerical counts, classifications, elemental composition, colour and

refractive index), and they are subject to various uncertainties. We denote the entire collection of observations on the trace by the multi-dimensional quantity Y. Observations are also made of corresponding properties of the control; these, also, are subject to uncertainty and we denote them collectively by X. We wish to use Y and X to address the following propositions that represent the prosecution and defence positions respectively:

H_p: the trace came from the same source as the control;
H_d: the trace came from some other, unknown, source.

Following Lindley and many others, our objective is to assign a value to the likelihood ratio LR which we write here as

$$\mathrm{LR} = \frac{f_{Y,X|H_p,I}(y, x|H_p, I)}{f_{Y,X|H_d,I}(y, x|H_d, I)}, \tag{1}$$

where $f_{\ldots}$ denotes a probability density function (PDF) and I denotes a framework of circumstantial information that is relevant to the consideration of the evidence. In a DNA profiling case, for example, it would be relevant to include in I evidence that the true source of the trace had a particular ethnic appearance. In a fingerprint case, it would be relevant to include in I that the trace was made by, for example, a right forefinger. For the rest of the paper we omit I from our equations, purely for brevity, though we recognize its importance at all times. Using standard properties of PDFs, we can write equation (1) as

$$\mathrm{LR} = \frac{f_{Y|X,H_p}(y|x, H_p)}{f_{Y|X,H_d}(y|x, H_d)} \frac{f_{X|H_p}(x|H_p)}{f_{X|H_d}(x|H_d)}. \tag{2}$$

We simplify this by making two assumptions.

Assumption 1. If H_d is true, then Y is independent of X:

$$f_{Y|X,H_d}(y|x, H_d) = f_{Y|H_d}(y|H_d).$$

There are situations where this assumption is invalid. In a DNA case, where the defence proposition is that the true offender is a close relation of the suspect, then the genotype X is clearly relevant to genotype Y. However, we accept it for the present context, recognizing that further studies may be directed to consideration of relatedness issues in fingerprint comparisons.

Assumption 2. X is independent of whether H_p or H_d is true:

$$f_{X|H_p}(x|H_p) = f_{X|H_d}(x|H_d).$$

Provided that y is absent from the conditioning, knowledge of which of the propositions is true does not influence uncertainty with regard to the observations on the control. It is difficult to visualize situations where this is not true. Given these two assumptions, equation (2) simplifies to

$$\mathrm{LR} = \frac{f_{Y|X,H_p}(y|x, H_p)}{f_{Y|H_d}(y|H_d)}. \tag{3}$$

In the case of DNA profiling, the denominator will be addressed by methods of population genetics. In other situations it would be customary to assign a value to the denominator by reference to a database of observations that have been made on samples taken from other sources and which is considered to be representative of the conditions that are specified under I and H_d. The issue of representativeness is, of course, important but it is too large for discussion here: useful aspects are covered by Aitken and Taroni (2004) and Champod *et al.* (2004). In any case, the control samples in the database should have been collected in a similar way to that followed

for the control sample in the case and the measurements should have been made by the same protocol on every control. Let there be N control samples in the database and denote the set of observations by $z_1, z_2, \ldots, z_N$. A value may be assigned to the denominator by, in principle, comparing the observations on the recovered sample with each member of the database in turn. Thus, we make the next assumption in lieu of an approximation.

Assumption 3.

$$f_{Y|H_d}(y|H_d) = \frac{1}{N}\sum_{i=1}^{N} f_{Y|Z,H_d}(y|z_i, H_d). \tag{4}$$

The quality of the approximation depends on the size and representativeness of the database.

Provided that the $\{z_i\}$ have been collected by the same process that led to x then it is reasonable to infer that the conditional PDFs for y are the same. It is convenient also at this stage to introduce simplified notation. Thus we write the next assumption.

Assumption 4.

$$\begin{aligned} &f_{Y|X,H_p}(y|x, H_p) = \pi(y|x), \\ &f_{Y|Z,H_d}(y|z_i, H_d) = \pi(y|z_i) \qquad \text{for all } i. \end{aligned} \tag{5}$$

Here $\pi(\cdot)$ represents a PDF that characterizes the uncertainty in a set of observations made on a trace, conditioned on a set of observations made on a control, given that the trace and control have come from the same source. It is no longer necessary to include H_p and H_d in the conditioning and so we arrive at the expression

$$\mathrm{LR} = N\,\pi(y|x) \Big/ \sum_{i=1}^{N} \pi(y|z_i). \tag{6}$$

We return to this after we have explained how the observations y, x and z_i are collected in the present context.

Finally we note that we present the results of our experiments as logarithms (base 10) of LR. An explanation of the notion of 'weight of evidence' as the logarithm of LR may be found in Good (1950).

4. Method for quantitative observations on fingerprints

When comparing fingerprints, examiners seek correspondences and discordances at all levels of detail between the mark and the print. At the *minutiae* level, the poor quality and the partial nature of the mark usually lead to the observation of less *minutiae* than in the print. Thus, the examiner searches the print for *minutiae* that correspond to the *minutiae* that are observed in the mark, attempting to create a set of paired corresponding *minutiae*. If there were discordant *minutiae* in either the mark or the print they would be noted; in general, these would imply a very small numerator. We do not consider such *minutiae* in this research: we consider only those comparisons where there are no discordant *minutiae*.

In the following sections, we used the method that was described by Neumann *et al.* (2007) for acquiring and organizing quantitative information from sets of *minutiae*; however, we modified the original metric.

4.1. Acquisition of minutiae features

The method that was described by Neumann *et al.* (2007) proposes a method for defining a *minutia* configuration as a polygon, through a set of triangles, providing a structure that is

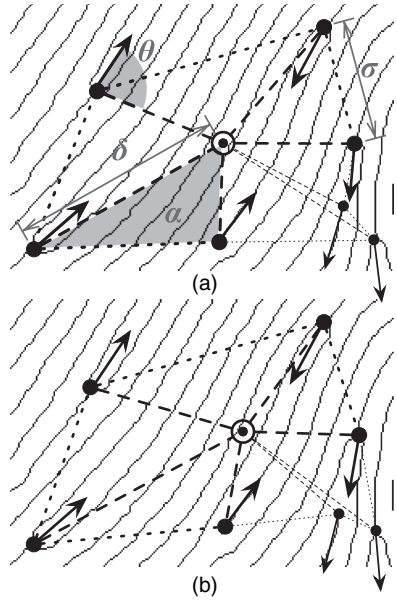


Fig. 2. Illustrations of the radial triangulation used to order *minutiae* (the variables listed in Table 1 are indicated in grey in (a); →, 'directions' of the *minutiae*): the type of the *minutia* is not indicated since it is a discrete variable; depending on distortion, radial triangulation of *minutiae* configurations may produce different ordered sets; the differences between (a) and (b) are presented by using thinner lines

unique, rotation invariant and robust to fingerprint distortion. For a given configuration the method requires the determination of a unique centre for the polygon, defined by the arithmetic mean of the Cartesian co-ordinates of the *minutiae*. The polygon is then defined by connecting the *minutiae*, which become the vertices, in clockwise order with respect to the centre of the polygon (Fig. 2). Each *minutia* is connected to two contiguous *minutiae* and also to the centre of the polygon, hence creating a series of contiguous triangles, which all share one vertex located in the centre of the polygon. Provided that the *minutiae* are always considered in the same order with respect to the centre, the capture of information for a given configuration is invariant to the orientation of the fingerprint on the image. Furthermore, the circular acquisition of the *minutiae* renders the acquisition process independent of the choice of starting *minutia*. Table 1 provides the notation for the data that are extracted from the *minutiae*. Note that the directions of the *minutiae* are recorded along the supporting ridge for ridge ending, and along the bisector of the splitting ridges for a bifurcation (Fig. 2).

4.2. Organization of minutiae features

For a configuration of k *minutiae* there are $5k$ features that are recorded in a feature array. A feature array that is recorded from a randomly selected *minutia*, which is subsequently denoted 1, may be written (following the notation in Table 1)

$$w^{(k,1)} = (\{\delta_i, \sigma_i, \theta_i, \alpha_i, \tau_i, \}, i = 1, 2, \ldots, k).$$

We use w in this context because we may be considering a mark y, print x, database member z or a simulated mark (for the notation, see later). Note that $w^{(k,1)}$ is an ordered set. The array that would be created by starting at the next (clockwise) *minutia*, which is denoted 2, would be

$$w^{(k,2)} = (\{\delta_i, \sigma_i, \theta_i, \alpha_i, \tau_i, \}, i = 2, 3, \ldots, k, 1).$$

Table 1. Data extracted from a *minutiae* configuration

Notation	*Description*	*Units*
δ	Radius—the distance between a *minutia* and the centre of the polygon	Pixels
σ	Side length—the distance between a *minutia* and the next contiguous *minutia* in a clockwise direction	Pixels
θ	Angle—the angle between the direction of a *minutia* and the line from the centre of the polygon to the *minutia*	Radians
α	Area—the area of the triangle constituted by a *minutia*, the next contiguous *minutia* and the centre of the polygon	Pixels
τ	Type—type of the *minutia* {ridge ending, bifurcation, unknown}	[0, 1, 2]

The definition of a distance between configurations requires that the starting *minutia* be specified. More generally, we shall use $w^{(k)}$ to denote the full set of arrays $\{w^{(k,i)}, i = 1, 2, \ldots, k\}$ for a given configuration. We use a subscript to refer to the feature array of a particular configuration, e.g. $w_r^{(k)}$.

4.3. Comparison of configurations

Consider two separate configurations, both of size k. One of these has been selected from a partial and degraded mark and one from a control print. The comparison of the quantitative information extracted from the two k-configurations can be visualized as the overlaying of the centres of the two configurations and the rotation of one of them, until the distance (which is defined below) between them is minimized.

Having established the notation in the previous section, the distance between the feature arrays $w_1^{(k)}$ and $w_2^{(k)}$ from the two configurations 1 and 2 is defined as follows.

First, we write $d\{w_1(\delta_i), w_2(\delta_j)\}$ to denote the difference between the radius measurements for *minutia* i in configuration 1 and *minutia* j in configuration 2. We denote the differences between the respective side length and area measurements similarly: $d\{w_1(\sigma_i), w_2(\sigma_j)\}$ and $d\{w_1(\alpha_i), w_2(\alpha_j)\}$. The angular difference is denoted $d\{w_1(\theta_i), w_2(\theta_j)\}$ and it is computed so that $0 \leqslant d\{w_1(\theta_i), w_2(\theta_j)\} \leqslant \pi$. The difference between the types of the selected pair is denoted $d\{w_1(\tau_i), w_2(\tau_j)\}$ and this takes the value of 0 if they are the same type, or 1, if they are different. The total difference between the two *minutiae* is the weighted arithmetic sum of the squares of the five individual terms:

$$\Delta_{1,2}^{(k,i),(k,j)} = q_\delta\, d\{w_1(\delta_i), w_2(\delta_j)\}^2 + q_\sigma\, d\{w_1(\sigma_i), w_2(\sigma_j)\}^2 + q_\alpha\, d\{w_1(\alpha_i), w_2(\alpha_j)\}^2 + q_\theta\, d\{w_1(\theta_i), w_2(\theta_j)\}^2 + q_\tau\, d\{w_1(\tau_i), w_2(\tau_j)\}^2. \quad (7)$$

The $q_.$-terms are simple multipliers, whose values are chosen at the optimization stage.

We pick a *minutia* in configuration 1 and, without loss of generality, call it *minutia* 1. Then there are k possible comparisons with configuration 2, depending on how the latter is rotated with respect to the former. If we denote the starting *minutia* in configuration 2 with i, where $i = 1, 2, \ldots, k$ then for each of these we calculated the combined distance, which is defined as

$$\Delta_{1,2}^{(k),i} = \sum_{j=1}^{k} \Delta_{1,2}^{(k,\,j),\,(k,\,i\sim j)}, \tag{8}$$

where $i \sim j$ denotes a cyclical offset. For example, if $k = 5$ and $i = 3$ then, as j counts 1, 2, 3, 4, 5, $i \sim j$ counts 3, 4, 5, 1, 2.

The distance between the two configurations is taken as the minimum value of $\Delta_{1,2}^{(k),i}$ over all values of i.

We give a brief explanation of the optimization of the metric in Section 6.

4.3.1. Switch in the positions of minutiae within an ordered set of minutiae

We recognize that, in some situations, two *minutiae* may be aligned with the centre of the polygon. Different distortions of the finger pad when leaving the print may produce different ordered sets of *minutiae* for the same configuration (Fig. 2). In this study, when the value of the angle subtended between two *minutiae* at the centre is less than $\pi/9$, we have noted that they may switch order under certain distortion conditions. In this situation, the feature arrays for both possible orders of *minutiae* in the configuration are recorded. When this configuration is compared with another, both distances are calculated but only the smaller is retained. Similarly, when multiple *minutiae*, or groups of *minutiae*, within a configuration can switch order, feature vectors for all possible switched orders are recorded. The distance between two sets of *minutiae*, which have groups of *minutiae* that can switch order, is the smallest possible distance between all respective feature vectors.

5. Databases used in this study

The model has been developed, optimized and tested by using several databases. Each database comprises fingerprint images captured digitally at a 1:1 magnification and at a resolution of 500 pixels per inch. The following data were extracted from every fingerprint that was used in the study:

(a) the finger of origin of the print, from 1 (right thumb) to 10 (left little finger);
(b) the general pattern of the friction ridge flow—these are standard categories, universally adopted within the fingerprints field, arch, loop and whorl;
(c) the Cartesian co-ordinates of each *minutia* in pixels, using the bottom left of the image as the origin;
(d) the direction of each *minutia* anticlockwise in radians, using the bottom left of the image as the origin;
(e) the type of each *minutia*—ridge ending, bifurcation or unknown;
(f) the Cartesian co-ordinates of particular features, such as cores and deltas in the ridge flow (Fig. 1), in pixels, using the bottom left of the image as the origin.

All fingerprint features were either labelled manually or determined automatically and verified manually. The data are used to create the feature arrays that have been described in the previous section. The databases are as follows.

5.1. Database A: reference database

The reference database, collected from a set of control fingerprint forms at the University of Lausanne (Switzerland), was used to support the assignment of PDFs in the denominator of LR in equation (8) (below). It consists of 12096 fingers from approximately 12000 individuals.

5.2. Database B: user influence

The model relies on the manual selection of fingerprint features by trained fingerprint examiners. 20 pairs of fingerprints (mark and corresponding print) were chosen from a Forensic Science Service fingerprint research data set and presented to 17 examiners. The selection process involves the display of each fingerprint image on a computer screen and the labelling of the *minutiae* by examiners. Our intention was to capture the reproducibility of the labelling and not to study whether a particular *minutia* was labelled or not. Thus, for each image, a template, indicating to the examiner which *minutiae* were to be labelled, was also displayed to the examiner during the annotation process. The pairs were marked up with various levels of repetition.

Six pairs of fingerprints, totalling 98 pairs of corresponding *minutiae*, were labelled by all 17 examiners. A model of the influence of the examiners was developed on the basis of these data. The model has different parameters for the variability of the labelling in marks and prints.

The complete data set of 20 pairs was used for optimization of the weighting function ψ that is described in Section 8.

5.3. Database C: fingerprint distortion

Individuals leave finger impressions in various conditions of distortion and pressure. Fingerprint examiners usually compare a partial, distorted and degraded mark with a single exemplar of a print that is collected under controlled conditions. Ideally, one would acquire multiple known impressions from a given finger under various controlled distortion and pressure conditions; however, this is unrealistic in casework conditions, if at all allowed under present legislation. Thus, a mathematical model, based on that described by Bookstein (1989), was developed to represent the distortion of finger pads when pressed on flat surfaces. This distortion model is used to generate distorted pseudomarks from the control print, or from prints in the reference database (see Section 7). It was developed by using 3520 images from five different fingers from five different donors (704 images for each finger). The model was tested by using a data set of 1100 images from 25 fingers from 16 donors (44 images for each finger). All images have been acquired under standardized conditions of pressure and distortion at the University of Lausanne, on behalf of the Forensic Science Service.

The model obviously does not cover all possible distortion and pressure conditions. Nevertheless, we propose it as a viable alternative—in normal pressure and distortion conditions—to the systematic collection of multiple control prints for each individual.

5.4. Database D: optimization

Parameters of the model were optimized by using a database obtained from the National Institute of Standards and Technology, USA (special database 27). It contains 258 mark–print pairs: in each pair, the mark has been recovered from a crime scene and the print has been considered by an examiner to be a match. Corresponding *minutiae* between the mark and the print in every pair were labelled and verified by fingerprint examiners.

5.5. Database E: validation

Having optimized and measured the performance of the model by using marks and prints originating from known sources, we performed a validation study using casework data. For this, a series of 364 marks collected from crime scenes and the corresponding control prints from known individuals, as identified by trained fingerprint examiners, were collected from closed case files from years 2003–2006, belonging to the North-Wales Police. The marks that were used

for the study had all been lifted by means of powder and tape. They were scanned at a resolution of 500 dots per inch and a magnification of 1:1.

To test the system under the most difficult conditions, the marks, which had anything between five and 36 *minutiae* recorded, were then encoded and searched against the national fingerprint database of the USA, using the integrated automatic fingerprint identification system. For each mark, the most similar control print in the database was retrieved. In each case, it was independently verified that the donor of the reference print retrieved from the US database could not possibly have left the mark at the UK crime scene. For each mark, we therefore obtained a triplet of data: mark—corresponding print from the UK—most similar non-corresponding print from the USA. The results are described in the validation experiments in Section 9.

6. Optimization of the metric

The optimization of the metric that was described in Section 4 was carried out in two stages. First, because the different variables have different units and ranges of magnitude, some kind of normalization was necessary. Second, because some variables may be more discriminating than others, the discriminating power of the metric was maximized by determining the optimal value of q for each variable. *Minutiae* configurations sampled from the marks and prints from the National Institute of Standards and Technology database D were used.

For this, each mark–print pair in the database was sampled to generate as many matching k-configurations as possible, for $k = 3$–12. Sampling rules were applied to restrict overlap between successive pairs (i.e. configurations of three *minutiae* were not allowed to share more than two *minutiae*). This process yielded, for example, 1384 paired configurations of three *minutiae* and 474 of 12 *minutiae*.

The optimization process is based on the notion that comparisons between corresponding *minutiae* configurations in mark–print pairs from the same finger (within source) should tend to yield small distances and comparisons between mark–print pairs from different fingers (between source) should tend to yield large distances. Thus, we cross-compared all configurations and measured their distance by using the metric. This produced, for example for $k = 3$, a distribution of 1384 within-source distances and a distribution of $\binom{1384}{2}$ between-source distances.

We found that, for each of the variables in Table 1, there was a linear relationship between k and the 95th percentile of the between-source distribution. This led us to normalize all variables to a corresponding range of magnitude by means of simple linear regression.

Secondly, we determined the relative contribution of each variable to the metric by considering the separation between the within- and between-source distributions. This was a largely heuristic process. Weights were assigned to each of the variables and the separation between the within- and between-source distribution was measured in terms of false acceptance and false rejection rates. These rates were measured by standard biometric performance assessment tools (Mansfield and Wayman, 2002; Gonzalez-Rodriguez *et al.*, 2006). The weights were updated to minimize both rates.

During this process, it was found that the area variable contributed little to discrimination and so we assigned the value 0 to q_α in equation (7), thus reducing the overall dimensionality of the information that was extracted from fingerprint configurations to $4k$.

7. Simulation of crime scene marks from control prints

In general casework, it is reasonable to expect that control prints are taken from suspects under circumstances that can be standardized to reduce the variability between multiple prints

from a same finger to an acceptable level. More specifically, control prints are taken such that the distortion effects resulting from the creation of a two-dimensional image from a three-dimensional finger pad are minimized, or at least acquired in a reproducible and robust way. The robust acquisition of control prints is necessary for fingerprint examiners to perform their work reliably. It is normally achieved by taking impressions of each of an individual's 10 fingers. The ridge structure of each finger is captured by rolling the finger from one side of the fingerprint pad to the other. This usually results in fingerprint images of high quality and contrast, with minimal influence from the support and the fingerprinting method. Usually, only one control print per finger is acquired, unless the first attempt results in a bad quality print, in which case, a second impression is collected.

Things are quite different, not surprisingly, when it comes to a mark left at the scene of a crime. It will, in most cases, be only a partial representation of the finger that it comes from. It will be subject to interference from the substrate on which it is laid. Dirt, blood, substrate pattern (wood, for example), shape (a bottle, for example) will all affect the information content of the mark. Also, in general, the mark will be subject to uncontrolled and potentially severe distortion from the way that it was laid down.

Making allowance for these sources of uncertainty is not helped by the fact that the crime mark is compared only against a single print from each finger of a possible suspect. Ideally, of course, we would like to see a set of control prints made under conditions that reflect those under which the crime mark was laid down but this is never possible. Present practice is for fingerprint examiners to make allowance subjectively for all the sources of uncertainty that pertain in a given case. In the approach that is presented here, we employ simulation to generate pseudomarks from reference prints.

Two mathematical models have been developed, which take account of the two main sources of variability that can be observed in casework conditions during the comparison between a print and a mark: variations between examiners in observing and labelling up features on marks and prints, and finger distortion.

7.1. Examiner influence model

The fingerprint comparison process consists of the comparisons between *minutiae* detected (and labelled) independently on the prints and on the marks. Because of the width of friction ridges and the elasticity of skin, there is an inherent variability in the determination of the exact position of any given *minutia* in a fingerprint. Furthermore, the generally poorer quality of the marks renders the labelling of the *minutiae* more difficult and thus subject to greater uncertainty than on the prints. This is independent of whether the location of a *minutia* is determined and labelled by a human or automatically. Nevertheless, the variability between human beings, and even within any particular individual, is assumed to be larger than the variability of automatic systems.

A model has been designed to replicate the variability of examiners when labelling *minutiae* in fingerprint images displayed on a computer screen. This was done by using database B to study and model the distribution of location and direction assigned to *minutiae* in repetitive labelling by trained fingerprint examiners. As expected, a larger variability was observed for the labelling of *minutiae* present on marks. Therefore two sets of parameters were set: one for marks and one for prints.

7.2. Distortion model

Distortion occurs when a finger pad is pressed against a surface. The pressure distorts the skin of the finger, thus affecting simultaneously all of the friction ridges that are present on the pad. The

model mimics the elasticity and the distortion of the skin of finger pads when they are pressed against a flat surface. It is based on thin plate splines (Bookstein, 1989) and was developed by using data extracted from database C. Sets of bending energy matrices were estimated from the training set in data set C. Once trained, the model could be used to create distorted sets of features from sets of features labelled on control prints. From one set of *minutiae* labelled on a control print, large numbers of marks distorted in 11 different directions (one central, eight linear and two torsional) may be generated. When variability of the labelling (as described in the previous section) is taken into account, a variety of sets of distorted configurations is generated.

The model incorporates an average severity of distortion that is equal for all control prints. A parametric model, which would be dependent on intrinsic characteristics of the fingers being distorted, may be considered in the future. Furthermore, a future version of the model will also need to consider the distortion of fingers when applied to surfaces that are not flat.

7.3. Generation of pseudomarks from prints

The design of the process for generating pseudomarks from control prints is based on our understanding of the deposition, observation and comparison of fingerprint features in practice. We start with the same assumption that fingerprint examiners make—that the control print is a near perfect representation of the friction ridge details on the finger pad. Then the mark is considered as a partial, distorted and degraded representation of the print. Furthermore, the process of observation introduces additional variability. So, given a particular configuration in a print, pseudomarks are generated as follows.

(a) The examiner influence model is applied repeatedly to the configuration to simulate multiple possible labellings of the location and direction of each *minutia*.
(b) The distortion model is applied to each of those possible labellings. Each different labelling of the print configuration that is obtained in (a) results in a different set of distorted pseudomarks.
(c) The examiner influence model is now applied repeatedly to the *minutiae* in each of the pseudomarks. This second application of the model simulates the variability of human examiners in observing and labelling the location and direction of the *minutiae* on crime scene marks.

8. Development of the theory

The formulation in Section 3 is applicable, in principle, to any type of evidence where quantitative observations are possible, e.g. shoe and tool marks, handwriting, glass or fibres. Now we develop the approach for the particular nature of the observations that we have studied in relation to fingerprints and which we explained in Section 4. In this context, the trace becomes a mark recovered from the crime scene and the control becomes a print taken from a suspect. The dimensionality of the data is determined by the number of *minutiae*, k, recorded in the mark. We denote the vectors of observations by $y^{(k)}$ in the mark, $x^{(k)}$ in the print and $z_i^{(k)}$ in the ith member of a suitable reference database of prints. In this study, we consider values of k from 3 to 12. There is an additional issue that we must introduce at this point. We have proceeded thus far as though the print contains a single k-configuration but this is not so. In practice, the print will contain many more *minutiae*, but an examiner will have selected a configuration of k *minutiae* in the print that appears to resemble the configuration in the mark. We now make a further assumption.

Assumption 5. The k-configuration in the print that is selected by the examiner as closest to the mark is the configuration that has minimum distance from the mark.

We denote the observations on this selected configuration in the print by $x_{\min}^{(k)}$. Then we consider the propositions

H_{p}: the mark was made by the finger that made the print,
H_{d}: the mark was made by some other, unknown, finger.

When we address the database, we look for the k-configuration in each of the prints that minimizes the distance from the mark. Denote that configuration for the ith member as $z_{i,\min}^{(k)}$. Then LR (6) takes the form

$$\mathrm{LR} = N\,\pi(y^{(k)}|x_{\min}^{(k)}) \Big/ \sum_{i=1}^{N} \pi(y^{(k)}|z_{i,\min}^{(k)}). \tag{9}$$

It is difficult to conceive of any situation in forensic science where, for a given item of evidence and a given pair of propositions, there is a single, uniquely defined LR. Inevitably, LR will be a function of the data summary that derives from the necessarily selective observations that are made on the evidence and also on the underlying model, that is inevitably founded on a set of assumptions. It then becomes necessary to establish a model-based method for computation of LR that satisfies desirable properties, given the nature of the *milieu*. Given the complexity of the data, it is clear that analytical solutions are not feasible, so our evaluation of equation (9) is based on Monte Carlo integration, as follows. For the numerator, we use $x_{\min}^{(k)}$ to create a set of simulated pseudomarks $\{\xi_1^{(k)}, \xi_2^{(k)}, \ldots, \xi_M^{(k)}\}$ that are generated as described in Section 7 to take account of the sources of variation to which observations on a crime mark are subject, i.e. user variability and distortion. Following the method that was described in Section 4, we calculate a set of distances $\{d_1^{(k)}, d_2^{(k)}, \ldots, d_M^{(k)}\}$ between the crime mark $y^{(k)}$ and each of the simulated marks. Likewise, for the denominator, we consider each member i of the database in turn, find the k-configuration $z_{i,\min}^{(k)}$ that is closest to $y^{(k)}$ and generate from it a set of simulated marks $S_M = \{\xi_{i1}^{(k)}, \xi_{i2}^{(k)}, \ldots, \xi_{iM}^{(k)}\}$. As with the numerator, we calculate the set of distances between these simulated marks and the crime mark: $\{d_{i1}^{(k)}, d_{i2}^{(k)}, \ldots, d_{iM}^{(k)}\}$. Then we assign values to the probability densities in the numerator and denominator by means of a suitably chosen function of the various distances, i.e. we calculate

$$\mathrm{LR} = N \sum_{j=1}^{M} \psi(d_j^{(k)}) \Big/ \sum_{i=1}^{N} \sum_{j=1}^{M} \psi(d_{ij}^{(k)}). \tag{10}$$

We have called ψ a 'weighting function' and we require some kind of inverse relationship between d and $\psi(d)$. To establish a precise form for ψ we have carried out experiments in which we generated sets of simulated cases where H_{p} was true and sets where H_{d} was true. We use $\mathrm{LR}(H_{\mathrm{p}})$ and $\mathrm{LR}(H_{\mathrm{d}})$ to denote LRs generated under these respective conditions. The form of $\psi(d)$ has been chosen to optimize the performance of the method with respect to the following *desiderata*.

(a) We require the distribution of $\mathrm{LR}(H_{\mathrm{p}})$ to be of predominantly large values. Values of less than 1 are undesirable.
(b) We require the distribution of $\mathrm{LR}(H_{\mathrm{d}})$ to be of predominantly small values. Values greater than 1 are undesirable.
(c) For any given mark-to-print comparison, we seek to minimize the variation in LR between different examiners.

(d) As the number of *minutiae* increases, the enhanced information content should be reflected in the distributions of $\mathrm{LR}(H_\mathrm{p})$ and $\mathrm{LR}(H_\mathrm{d})$.
(e) Infinite LRs are to be avoided: thus the denominator cannot become 0.

In addition to these requirements, we also require that ψ should take account of the intrinsic quality of the mark. The effect of the quality of the mark on the conclusion of a fingerprint examination is well known to fingerprint examiners and it is intuitively reasonable that configurations on more degraded marks will be more difficult to label. This will affect distances between the mark and the control print, especially if H_p is true. In our function, the effect of the quality of a mark on the labelling of its characteristics is summarized in a statistic $T^{(k)}$ that is computed for each mark. This is done by generating a set of 50 simulated marks from $y^{(k)}$ by using the examiner influence model. All $\binom{50}{2}$ distances between the simulated marks are calculated and $T^{(k)}$ is taken as the 95th percentile. This is only an early attempt at introducing the contribution of the quality of the mark to the model: refinement of the treatment is left for future work.

Database A was used as the reference database throughout the experiments; databases B and D were used for optimization. The first, based on 20 mark–print pairs that were marked up to various levels of replication by 17 different examiners, was used to optimize the weighting function with respect to variability between examiners; the second, based on a National Institute of Standards and Technology database of 258 pairs, was used to optimize with regard to the remaining *desiderata*. We first explored the possibility of using a simple step function:

$$\psi(d^{(k)}) = \begin{cases} 1 & \text{if } d^{(k)} \leqslant T^{(k)}; \\ 0 & \text{otherwise.} \end{cases} \tag{11}$$

This was inadequate in two ways. First, in those cases where H_p was true, the numerator of LR was oversensitive to between-examiner variation. Second, for marks displaying more than seven *minutiae*, none of the distances computed between the mark and the pseudomarks simulated from control prints from the reference database fell within $T^{(k)}$, so requirement (e), that of avoiding zero denominators, was not met.

We next investigated the performance of exponential weighting functions of the form

$$\psi(d^{(k)}) = \exp\left(-\frac{\lambda d^{(k)}}{T^{(k)}}\right). \tag{12}$$

Various experiments were carried out to find an acceptable value for λ.

The advantage that is gained from this function was that the sensitivity of the numerator of $\mathrm{LR}(H_\mathrm{p})$ to examiner variation was greatly reduced, compared with the simple threshold approach. It was also a partial solution to the problem of zero denominators. However, because of the rapid decay of the tails of the function the denominators were very sensitive to between-examiner variation.

Next, we investigated weighting functions based on the beta function, initialized at the value 1 for $d_0 = 0.0005$, where d_0 was the smallest value into which the distances were binned.

$$\psi(d^{(k)}) = B(d^{(k)}, \lambda T^{(k)}). \tag{13}$$

Because of the relatively rapid fall of the function over relatively small distances, the problem of unacceptable between-examiner variation in the numerator of $\mathrm{LR}(H_\mathrm{p})$ returned.

In summary, the exponential function provides good performance in the numerator of $\mathrm{LR}(H_\mathrm{p})$ but poor performance in the denominator of both $\mathrm{LR}(H_\mathrm{p})$ and $\mathrm{LR}(H_\mathrm{d})$ in relation to examiner variability. In contrast, the beta function provides good performance in the denominator but poor performance in the numerator of $\mathrm{LR}(H_\mathrm{p})$. Furthermore, the problem of meeting *desideratum* (d) remains. To address this issue, we note that the exponential function dominates over

small distances and should incorporate $T^{(k)}$. The beta function dominates over large distances when $T^{(k)}$ is not relevant. So we replaced the second beta parameter with a function of k:

$$\psi(d^{(k)}) = \exp\left(-\frac{\lambda_1 d^{(k)}}{T^{(k)}}\right) + \frac{B(d^{(k)}, \lambda_2 k)}{B(d_0, \lambda_2 k)}. \tag{14}$$

The coefficients λ_1 and λ_2 were optimized by using databases B and D to satisfy the *desiderata*.

We are aware that we might be open to criticisms of arbitrariness in our choice of weighting function. Indeed, the potential range of choices is infinite, but we stress that our selection has been dominated by the *desiderata* that we have clearly stated. The performance of the method may be judged from the validation experiments that we present in the next section.

9. Validation of the method

Two validation experiments were performed using database E (Section 5.5). As we have explained, there were 364 triplets, each of which consisting of a UK crime mark, the control print that the North-Wales Police fingerprint examiners considered to be from the finger that made the mark, which we call the 'UK print', and the closest print retrieved from the US national database, which we call the 'US print'. For each triplet, we accept the UK mark as being from the same finger as the UK print (H_p true), and the US print as being from some finger other than that which made the UK mark (H_d true). The number of *minutiae* recorded in each comparison varied from 5 to 36.

The search algorithm of the integrated automatic fingerprint identification system works on completely different principles from those that we employ for assessment of evidential weight. We also realize that some of the retrieved close non-matches may not challenge human fingerprint examiners. Nevertheless, the search algorithm relies on the same fingerprint features, such as the general pattern of the ridge flow, and *minutiae* location and direction. Furthermore, current integrated automatic fingerprint identification system algorithms are very accurate in searching crime marks and retrieving reference prints from large databases on the basis of their similarity (Indovina *et al.*, 2009). Thus, we are satisfied that this experiment is the most convenient way of retrieving challenging data for testing the model.

9.1. Small experiment

For the small experiment, we considered only those triplets where the mark comprised between five and 12 *minutiae*. This represented 122 triplets and their breakdown by number of *minutiae* is given in Table 2. In each triplet, there was a comparison where H_p was known to be true and a comparison where H_d was known to be true: continuing the notation from the previous section, these yielded values of $\mathrm{LR}(H_p)$ and $\mathrm{LR}(H_d)$ respectively. Our requirement of the approach is that values of the former tend to be large and that values of the latter tend to be small. The results are summarized in Fig. 3.

Desiderata (a), (b) and (d) from Section 8 are clearly satisfied. There is a clear separation between the LRs calculated for $\mathrm{LR}(H_p)$ and $\mathrm{LR}(H_d)$, and this can be observed for all k. The

Table 2. Breakdown, by number of *minutiae*, of the UK marks searched against the US national database

Number of *minutiae*	5	6	7	8	9	10	11	12	more than 12
Number of marks	1	7	12	14	18	18	35	17	242

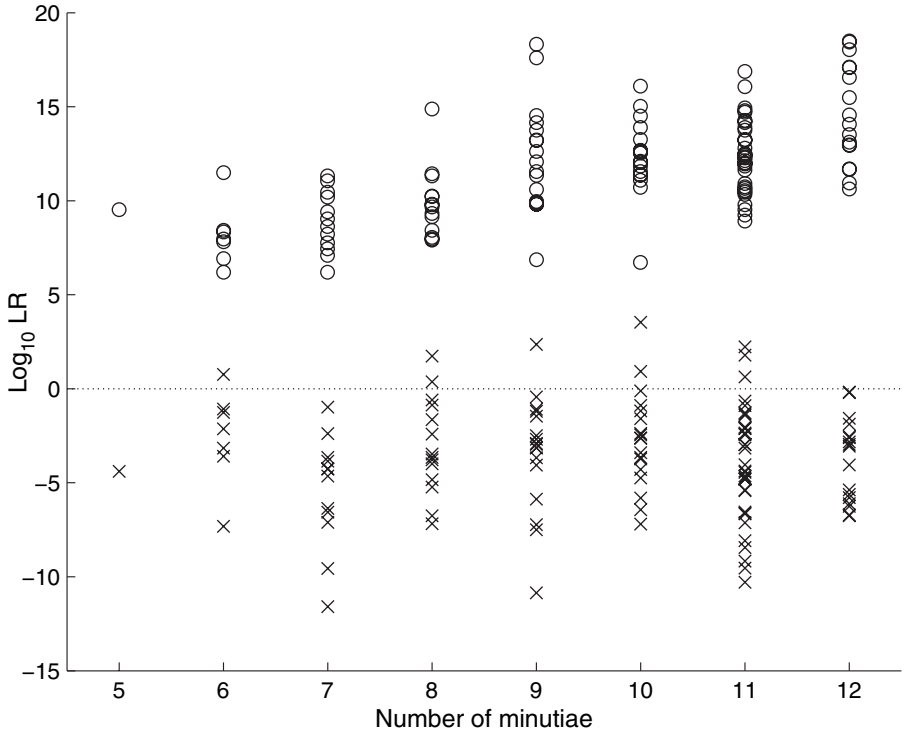


Fig. 3. LRs calculated from 122 mark–print–print triplets in the small validation experiment

few occasions when $\mathrm{LR}(H_\mathrm{d})$ exceed 1 are to be expected, particularly when it is borne in mind that the prints have been retrieved as the closest among 600 million.

The separate examination of the numerators and denominators of $\mathrm{LR}(H_\mathrm{p})$ and $\mathrm{LR}(H_\mathrm{d})$ in Fig. 4 enables us to explore further the performance of the model. Fig. 4(a) shows the numerators when H_p is true—num(H_p)—and when H_d is true—num(H_d). The denominators—den—for all of the marks (these are the same for both $\mathrm{LR}(H_\mathrm{p})$ and $\mathrm{LR}(H_\mathrm{d})$ since they are only conditioned on the mark) are shown in Fig. 4(b). Fig. 4(a) shows that, for the cases where H_p is true, the numerators fall below 10^{-5} only occasionally. From detailed inspection of such instances we find that they are occasions where either there was exceptional distortion in the mark, or where the labelling of a *minutia* was not as accurate as it should have been. In contrast, the numerators fall below 10^{-5} for most of the cases when H_d is true. From the comparison of the range of values of num(H_d) in Fig. 4(a) with the range of values of den in Fig. 4(b), we see that instances of $\mathrm{LR}(H_\mathrm{d}) > 1$ are caused by the division of very small values of num(H_d) by similarly small values of den.

This observation highlights a potential contribution of the model to quality assurance of fingerprint examination. In any case where LR is calculated to be greater than 1 and the numerator is smaller than 10^{-5} (or, perhaps, some other guideline, determined from an operational trial) the model would call for the attention of the examiner. In our experiment (and assuming a quality threshold of 10^{-5}), six out of the 122 cases where H_p was true, and nine out of the 122 cases when H_d was true, would have been called for review. All the observed cases where $\mathrm{LR}(H_\mathrm{d}) > 1$ would therefore have been reviewed. For example, the largest value of $\mathrm{LR}(H_\mathrm{d})$ was 3400 (approximately) in a comparison based on 10 *minutiae* (Fig. 3). Inspection shows that this

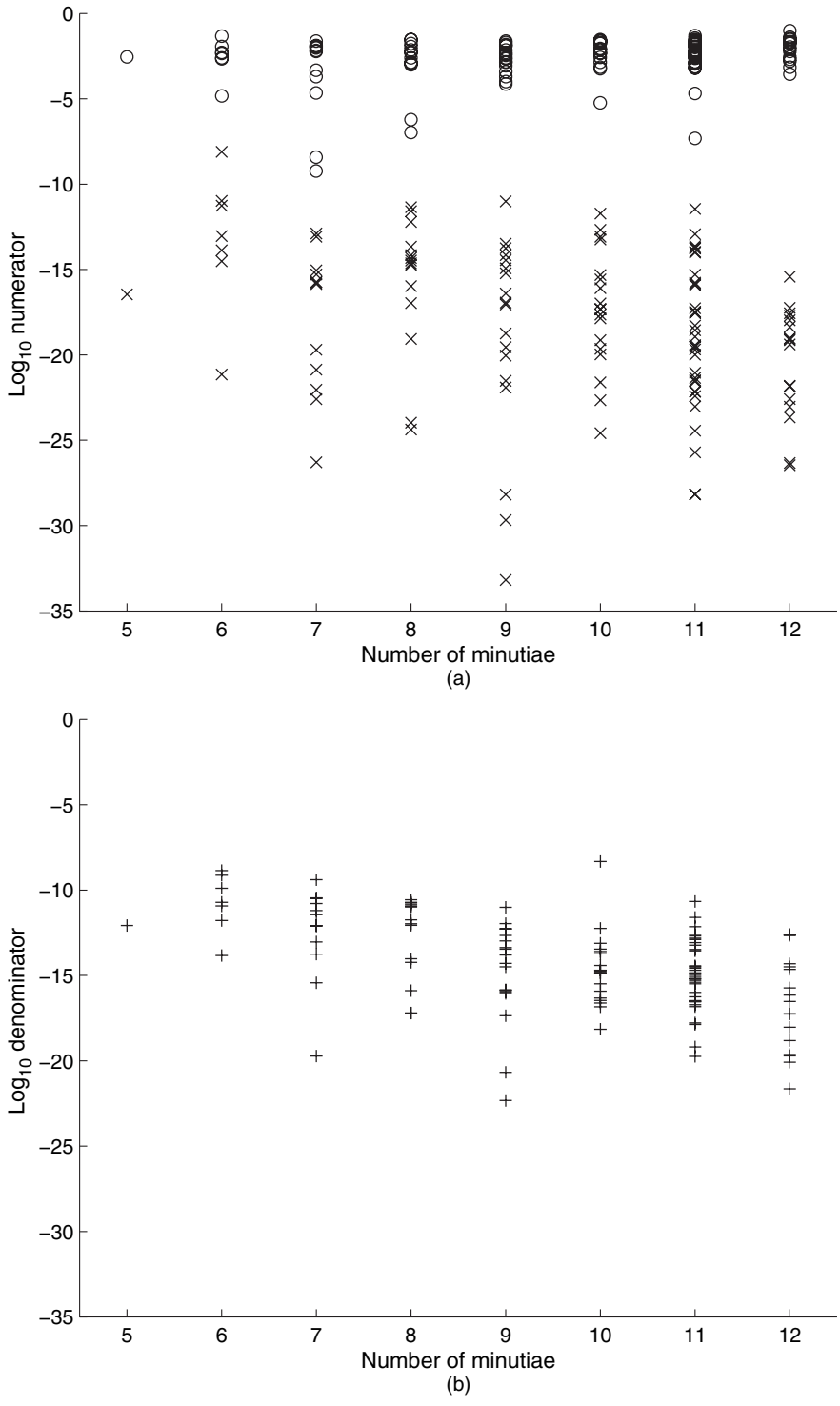


Fig. 4. Breakdown of the LRs reported in Fig. 3 (○, num(H_p); ×, num(H_d); +, den): (a) numerators; (b) denominators

arose from the division of a numerator of 8.25×10^{-14} by a denominator of 2.24×10^{-17}. The magnitude of the numerator would clearly send a quality warning to the fingerprint examiner.

The US prints have, of course, been selected and compared with the marks automatically. In casework, the mark and print would have been compared by an examiner before being entered into the system for numerical assessment of evidential weight. Examiners would use the model only if they could not exclude the source of the known print as the source of the crime mark. Hence, it is unlikely that a comparison that had been first inspected by an examiner would subsequently yield a small numerator. If this did occur, we would expect the comparison to be referred for further study so that the examiner might consider a wrong labelling of one, or more, *minutiae* an unusual distortion of the mark, or simply that H_d is true.

9.2. Large experiment

The data for the large experiment were generated by resampling k-configurations from the 364 UK marks for $k = 3$–12 and selecting comparison configurations from the corresponding UK and US prints. For this experiment only, we denote the total number of *minutiae* observed on a mark in a given triplet by k'; then the sampling proceeded as follows.

(a) Provided that $k' \geqslant k$, a set of k was selected subject to the constraints that the k *minutiae* had a direct spatial relationship and no other *minutiae* were contained within the polygon defined by the k *minutiae*.
(b) The corresponding k-configuration was selected on the UK print.
(c) The corresponding US print was searched for the k-configuration that had the minimum distance from the mark configuration.
(d) The process was repeated until all possible k-configurations had been selected for the given mark given the restriction that any new configuration could not share more than h *minutiae* with any configuration that had already been sampled. The value of h was changed with changing k (Table 3).

The numbers of k-configuration triplets (for $k = 3$–12) thus generated are shown in Table 3. It is worth emphasizing that this experiment pushes the model hard. Not only has each US print been selected as the closest out of 600 million, but now we are repeatedly subsampling marks: whereas a k'-configuration in a mark might lead to a small LR when compared with the US print, subsampling smaller sets of size $k \leqslant k'$ within that configuration is expected to yield one or more subsets that yield larger LRs.

The results for $\mathrm{LR}(H_p)$ and $\mathrm{LR}(H_d)$ are shown for $k = 3$–12 in Fig. 5. The results of this experiment are consistent with those observed before. The separation between the two distributions is good for all values, particularly when we recall the point that was made in relation to the small experiment. In this large experiment, both mark and print configurations in each comparison were selected automatically: in practice, we would expect that an examiner would mark up each comparison manually before entering the system for calculating a numerical assessment. In this

Table 3. Breakdown, by number of *minutiae*, of the resampled triplets of *minutiae* configurations from the UK marks

k	3	4	5	6	7	8	9	10	11	12
h maximum shared *minutiae*	0	1	2	3	4	5	6	7	7	8
k-configuration triplets	1177	1317	1383	1343	1270	1192	1103	1008	568	486

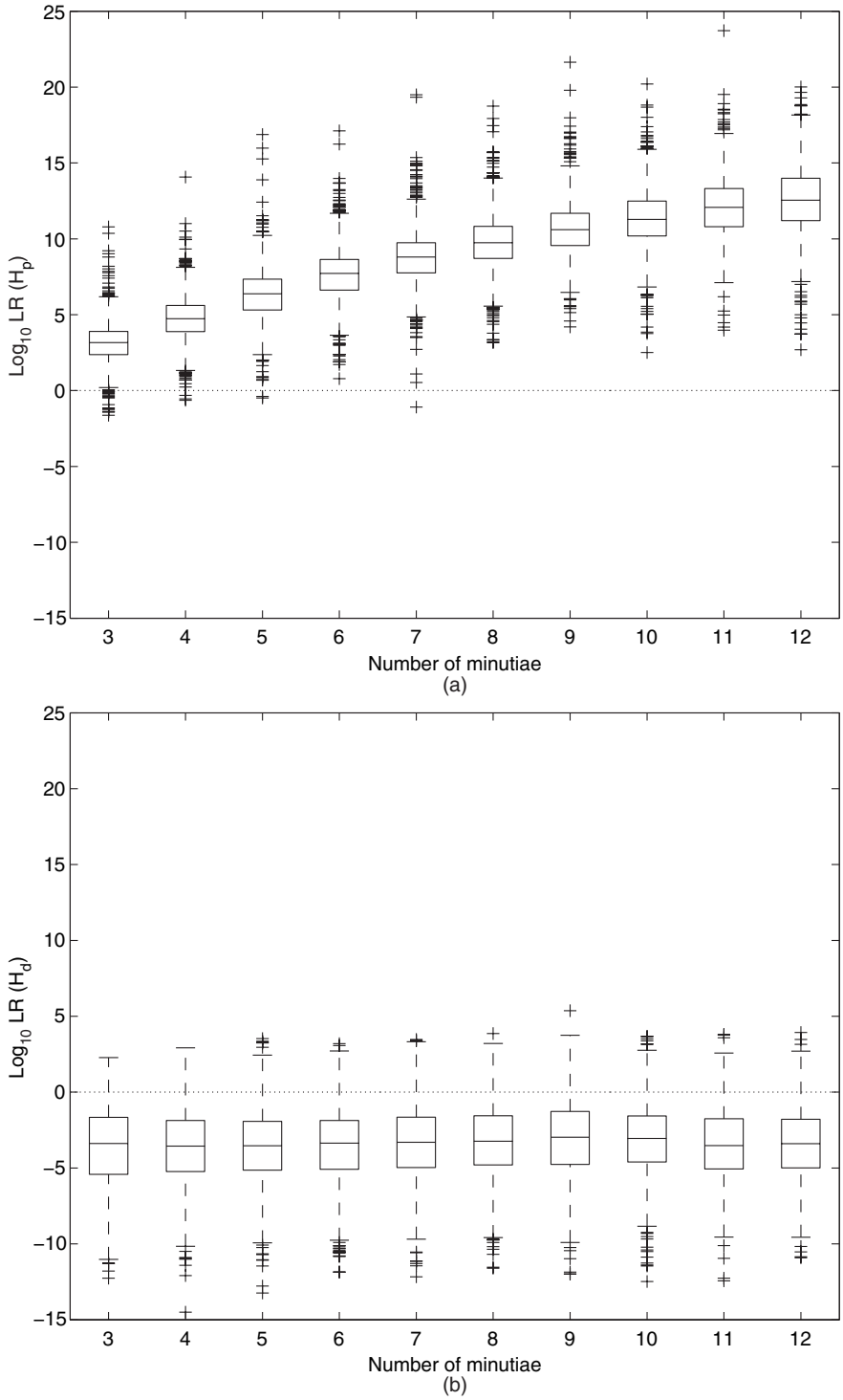


Fig. 5. Distributions for LR(H_p) and LR(H_d) from the large validation study, for configurations of sizes $k = 3$–12: (a) LRs when H_p is true; (b) LRs when H_d is true

way, we would expect that very few of the instances where $\mathrm{LR}(H_\mathrm{d})$ exceeded 1 would have been entered into the system. Even if they had, the use of the model as a quality assurance threshold, as discussed earlier, would prove useful to flag up those comparisons where the numerator value was very small. For instance, 888 out of a total 10847 LRs computed when H_d was true were greater than 1. Out of these 888 LRs, only 75 incorporated a numerator greater than 10^{-5}. All these 75 LRs were generated from configurations of $k = 3$ *minutiae*. Given the nature and the design of the experiment, this result is not surprising. Indeed, it is intuitively reasonable that the probability of finding, by chance, configurations similar to a given mark is higher when the number of *minutiae* is low.

9.3. Review of the experiments

The results from the two experiments are consistent with each other. In both experiments

(a) very few instances of $\mathrm{LR}(H_\mathrm{d}) > 1$ were observed and most of them would be filtered out because of the presence of a small numerator,
(b) even fewer instances of $\mathrm{LR}(H_\mathrm{p}) < 1$ are observed, most of them for configurations with lower numbers of *minutiae*, and, in all cases, low values for $\mathrm{LR}(H_\mathrm{p})$ are provoked by an extreme distortion of the mark or by a mislabelling of the mark or the print,
(c) the median value of $\mathrm{LR}(H_\mathrm{p})$ increases with k and
(d) for each value of k the values of $\mathrm{LR}(H_\mathrm{p})$ range over several orders of magnitude, and thus the distributions for different values of k overlap each other.

Observations (a) and (b) indicate that the probability of misleading evidence, either in favour of the prosecution or the defence, is very low, especially when considering that the model was tested under extreme circumstances, using a selection of close non-matches out of a database of 600 million prints. The results were computed from automatically generated data, without the filtering that human examiners would have undoubtedly added. Our model takes into account only a small selection of the features that fingerprint examiners are using. We can safely assume that the use of a wider range of features by examiners would make their examination more discriminating, thus further reducing the probability of misleading evidence.

A final point can be made regarding the recurrent discussions in the fingerprints community about the use of a point count as a criterion for concluding the identity of the source. Observations (c) and (d) show that the point count is only one factor among several when it comes to assessing weight of evidence; this provides support for the school of thought which maintains that there is no statistical basis for specifying a numerical points standard.

10. Discussion of propositions—scope for future work

In this paper, we have confined our analysis to 'finger' propositions (Section 8). In casework, we would, in general, prefer to address 'person' propositions of the form

H_p: the mark was made by the person who provided the control prints,
H_d: the mark was made by some other, unknown, person.

The first point to recognize here is that every person who provides control prints to the police does so on a form, which is widely known as a '10-print form', that captures all of his or her fingers. This applies to the suspect in the case at hand and also to all those whose prints are kept on a central database. At the very simplest, we could allow for this by changing the definition for $Z^k_{i,\min}$ as the closest k-configuration across all the control prints from the ith member of the database.

In equation (10) then, N would represent the size of a database of 10-print records. For this study, we used a database of 12096 single fingers from approximately 12000 different individuals (database A). If we made the crude (and unexplored) assumption that the variation between fingers within an individual was the same as the variation between individuals then we could artificially reconstruct this as a database of approximately 1200 10-print records. Then the N in equation (10) would be reduced by a factor of 10 and all the LRs that were calculated and described in Section 9 reduced accordingly.

However, there are two issues that, subject to the availability of an appropriate reference database, would mitigate against such a reduction in the weight of evidence. The first is that often the fingerprint examiner will be able to make some sort of inference about which of the donor's 10 fingers will have left the mark. If the donor has held a glass, for example, then it will be natural for the thumb to be opposed to two or more other of the fingers and it would be expected that the forefinger was above the second finger, and so on. Overall, this is quite a complicated situation but it is believed that examiner inferences of finger number are quite reliable. So if, on the basis of a scene examination in a particular case, the examiner assigned a high probability to the proposition that the mark was left by a right forefinger then it would appear reasonable to address prosecution and defence propositions that specified that finger. The second issue is the pattern type of the mark and print. This paper is concerned with *minutiae* but it is the case that useful classification information derives from the broad pattern that the ridges in a given mark or print form. There are three main types: whorl, loop and arch. The comparison of these patterns between the mark and the print form an important part of the fingerprint examination process. Assigning such a pattern to a mark provides additional information that can be incorporated into the calculation of LR. This is something for further work.

A related matter is that of contiguous marks. In the example of a glass which we have just given it might be that there are *minutiae* configurations in two or more of the crime marks that can be compared with configurations in corresponding fingers of a suspect. There is clearly a potential for such information to enhance evidential value but, again, it is something for further work.

The consideration that our reference database consists of only 12 096 single fingers from Swiss donors (mostly white Caucasian males) means that further work is necessary to investigate representativeness and robustness in relation to general casework in a heterogeneous population. Future research needs to address issues such relatedness, as well as differences in the statistical behaviours of populations of donors of different sex and ethnicity. If relevant, the inclusion of sex and ethnicity information in the calculation of LR needs to be formally described following the work done for DNA evidence. The use of larger databases, such as national fingerprint databases, and ultimately the convergence of automatic fingerprint identification systems and fingerprint statistical models will be needed to perform those studies. This convergence will also offer some obvious benefits from the operational point of view.

11. Conclusion

We have explained a method for arriving at a quantitative assessment of the weight of evidence that is provided by a fingerprint comparison. We have shown how the method has been optimized in relation to *desiderata* that are appropriate for a criminal justice system. Our experiments have shown that the method performs well under conditions that are even more stringent than those that pertain in casework.

Within the limitations that we have expressed, such as representativeness of the reference database, we maintain that the model can realistically be employed to complement the existing

procedures for forming opinions. Our research represents an important contribution to the search for a scientific foundation for the evaluation of fingerprint evidence; in turn, it contributes to the goals of balance, robustness and transparency.

We do not pretend that statistical models are the answer to all the challenges that face the fingerprint community. The fingerprint profession has made much progress over recent years in development of quality management procedures, including recruitment, training and certification of fingerprint examiners. Nevertheless, it is clear that the provision of a quantitative method for assessing weight of evidence will provide a powerful agent for greater transparency in the selection, documentation and evaluation of ridge details, and for the convergence towards a common reporting standard. This, in turn, will contribute to the reduction of variability between individuals and organizations.

We have explained that potentially useful evidence must be discarded in the case where the examiner considers that the quality and quantity of ridge detail on the mark is insufficient to express a categorical opinion. Our method provides a means of dealing with such cases and reporting what may be considered as corroborative evidence. A field study is currently being carried out in the State of Minnesota (USA) to define more clearly the benefits and challenges of the use of the model in casework.

In the immediate future, we do not see that the current practice of presenting categorical opinions will change. But longer term we expect an evolution towards a framework that is similar to that which underpins DNA evidence. The pace of change is difficult to determine. The work that we describe here represents, we believe, an important contribution to that change. Already, it has formed the basis of training workshops in the UK, USA and in Europe. The response from practitioners has been cautious but predominantly favourable. Much work has been done by the Forensic Science Service in developing a computer-based platform for casework use. Ultimately, we see this kind of approach replacing the existing paradigm but there will be a substantial transition phase in which we see the process as offering improvements in quality assurance, training and calibration.

Still greater challenges are to be expected when the new paradigm is introduced to the courts. We have seen several years of courtroom battles in relation to DNA evidence and, in general, these have proved to be beneficial to the science. However, the general level of comprehension of the principles of evaluation of evidence among legal practitioners is very low and the notion of a quantitative measure of evidential weight involves subtle issues that are absent from a simple categoric opinion of the identity of the source. It remains to be seen how future battles play out but we see models such as ours as a powerful platform for change.

In the normal course of events, a paper such as this would be submitted for publication in the forensic science or legal literature. Nevertheless, when these methods come to court it is inevitable, as with DNA profiling, that the attention of statisticians will be brought to bear and so we believe that it is necessary to address that step now. We see this as a major step on the path to paradigm change in the operational world of fingerprint identification.

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Discussion on the paper by Neumann, Evett and Skerrett

C. G. G. Aitken (*University of Edinburgh*)

The authors are to be complimented on their work to bring the advantages of the likelihood ratio to evidence based on a comparison of fingerprints. For most of the 20th century fingerprints were the gold standard of evidence until the advent of DNA profiling and several high profile cases of mistaken identity based on fingerprint comparisons challenged this position.

In 1977, a fundamental paper by Dennis Lindley (Lindley, 1977) paved the way for the introduction of the likelihood ratio approach in general for the evaluation of evidence. The advent of DNA profiling in the mid-1980s brought probabilistic reasoning and likelihood ratios to the attention of the courts.

One of the people in the forefront of the development of the likelihood ratio for evaluation and interpretation of evidence is one of the authors of this paper. A large reason for the current recognition by the forensic science profession and by the legal world of the benefits of probabilistic reasoning is due to the work of Ian Evett. He has continuously challenged people to think logically about their processes for evaluation and interpretation of evidence and has developed scientific methods which take due account of the legal constraints. All of us working in this area owe Ian a big debt of thanks for leading the way so tirelessly for so long.

However, it should be noted that the Bayesian paradigm has not been well received in the USA. It has been commented to me that this is, in part at least, because of the paradigm's dependence on databases for the development of the models.

There are many factors to consider regarding databases—their reliability, their objectivity, their size, their relevance and their invariance to time, for example. The authors comment, rightly, that the issue of representativeness is 'too large for discussion' here. However, it is difficult to imagine the fingerprints of individuals changing over time in the way that the components of other databases may do. The authors are to be complimented on the care that they have taken with their choice of databases and with the size of them.

There is also the recent case involving a comparison of shoe marks, an obvious future extension of the work in this paper, of R. *versus* T. (2010, EWCA Crim 2439) in the England and Wales Court of Criminal Appeal where the use of likelihood ratios was rejected, partly because of the use of databases and arguments over their representative nature. The ruling prompted comments in the forensic scientific literature (Berger *et al.*, 2011) and the legal literature (Redmayne *et al.*, 2011). The Court was concerned that

> 'in the absence of hard data, numerical quantification could produce an almost fraudulent "verisimilitude of mathematical probability" ',

and that

> 'an approach based on mathematical calculations is only as good as the reliability of the data'.

What we have in this paper are 'hard data' and 'reliable data'. I wonder what the courts will then make of this application.

Presentation is another area in which the authors' view would be of interest. Verbal scales based on likelihood ratios exist and it was the scale that was used by the Forensic Science Service which was criticized by the Court in R. *versus* T. However, presentation of a numerical value does present a false impression of exactitude and the presentation of a confidence interval presents difficulties of interpretation. A verbal scale provides an interpretation which is relatively easy to understand by a layman and yet is not misleadingly precise.

There is still a place for personal experience. Redmayne *et al.* (2011) comment in the context of R. *versus* T. that

> 'When the medical expert witness testifies that I have never encountered such a case in forty years of clinical practice, he is utilising data but he calls them "experience", relying on memory rather than any formal database open to probabilistic calculations. It would be just as foolish to maintain that memory and experience are never superior to quantified probabilities in criminal litigation, as it would be to insist that memory and experience are always preferable to and should invariably displace empirical data and quantified probabilities in the courtroom.'

In a manner similar to that of the medical expert witness, the use of likelihood ratios in fingerprint examination must be tempered with the use of the experience of the examiners.

Before finishing, I wish to comment on research and forensic science. The Forensic Science Service, a service with which all three authors have or have had affiliations, is closing. It is not clear that those providers of forensic science services remaining will be able or willing to devote resources of a similar kind to those applied here to other similar important work. Following the closure of the Forensic Science Service and its research capabilities, I hope that some way can be found to enable research of the thoroughness and quality that has been shown here to be conducted.

There is a long way to go before the paradigm presented here will become as accepted in the courts as DNA profiling is but long journeys all start with small steps. This paper is a rather large small step. I have great pleasure in proposing the vote of thanks.

David J. Balding (*University College London*)

It is an example of the irrationality of much public discourse that the notion that fingerprints are unique has enjoyed prominence as a justification for categorical identification of an individual from a fingerprint mark. The notion is believable, but whether a given partial and distorted mark suffices unequivocally to eliminate all but one potential source is a different question, as the authors have so clearly described in their paper. The practice of not using an evidential mark unless the examiner feels able to declare certain identification is wasteful of evidence. The major potential benefit to society from this paper is that it is now possible to evaluate with sufficient rigour for use in court a large proportion of 'messy' marks that currently would be discarded as unusable. I congratulate the authors on their achievement.

The uniqueness claim for fingerprints was based on the statistical model of Galton (1892), of which Stigler (1995) commented '... with minor qualifications ... can be defended as rigorous and correct'. After playing a role in establishing the power of fingerprints, Galton's calculation was virtually forgotten and played no role in courtroom practice. It is astonishing that more than 100 years later no alternative statistical model has, until now, become established as underpinning the evidential value of fingerprints.

The acme of forensic science today is DNA profiles, which are similar to fingerprints in showing high interindividual variability, yet stability over lifespan and a substantial ability to withstand environmental insults. Fingerprints are less heritable than DNA and so have greater power to distinguish relatives. The authors did not discuss relatedness and ethnicity for fingerprints, which are of central importance for DNA evidence. They are less important for fingerprints but can they be safely neglected? Galton found ethnic differences in fingerprints, but perceived without formal argument that they were not 'significant', as we would now say. What is the current state of knowledge? Why did the authors not test their method against relative pairs or even different fingers from the same individual?

DNA profile analysis benefits from well-established theories of inheritance and evolution. For many areas of forensic science, there is little unifying theory, and conclusions are based largely on a collection of instances of multiple measurements of a trait obtained from the same source, compared with measurements from sources that are similar but different. Rutherford's infamous quip that science is either physics or stamp collecting was never accurate but had an element of truth at the time, and has some resonance in today's forensic science with DNA evidence taking the role of physics. This paper advances the claim of fingerprint analysis to have a rigorous theoretical basis, as the models of user variability and of distortion are susceptible to rigorous evaluation: not so for the arbitrary weighting function. The authors first consider a step function, and when that was found wanting they tried an exponential function. This seems a large jump in function space: if a step function was judged worthy of consideration, a logistic function seems a natural alternative.

Voice evidence has some similarities to fingerprints; there is scientific theory relating to auditory, phonetic and linguistic qualities of voice recordings, and a body of methods from signal processing, but a central problem is how best to make use of databases of same source and different source recordings. Research

in the field has focused on calibration of likelihood ratios by using ideas such as cross-entropy, and this may have some relevance to the authors' work (Brummer and du Preez, 2006; Gonzalez-Rodriguez *et al.*, 2007; Morrison, 2010).

In England and Wales, quantitative analysis of non-DNA forensic evidence currently lies in the shadow of the R. *versus* T. judgement (All ER (D) 240 (Oct); 2010, EWCA Crim 2439) that severely curtails efforts to establish a more rigorous scientific footing for evidence evaluation, as called for in a landmark US report (National Research Council of the National Academies, 2009). There were problems with the presentation of evidence to the court trying T., but the result of the sweeping judgement is that attempts by forensic scientists to evaluate evidence logically, making explicit their data and assumptions, are now ruled unacceptable; only DNA evidence is given special exemption. Otherwise courts insist on only hearing statements such as 'this mark could have come from this item', which are often close to meaningless (an open-minded scientist might regard a wide range of sources as satisfying the vague 'could have' criterion) and masks subjective assessments that can vary enormously among experts.

Whether this paper can help UK courts to overcome their apparent fear of data-based logical reasoning, time will tell. R. *versus* T. is a major setback, but the larger tide is in our favour. I congratulate the authors on making a major advance in putting fingerprint evidence on a more sound quantitative basis, and it gives me great pleasure to second the vote of thanks.

The vote of thanks was passed by acclamation.

Bernard Silverman (*Home Office, London*) (© Crown copyright 2012. Reproduced with the permission of Her Majesty's Stationery Office and the Home Office)

The problem and data that are discussed in this fascinating paper can be seen as an example of functional data (see, for example, Ramsay and Silverman (2005)) where the data are not in the form of the numbers or vectors of standard statistical data but are of some more complex form. Within the functional data framework, the *minutiae* are 'landmarks' within the observed marks. If we consider landmarks then we are led into the areas of spatial statistics and the statistics of shape, and these will contain insights which will be pertinent in the future research which I very much hope this paper will provoke. Some immediate questions which suggest themselves are as follows. What is the appropriate choice of landmarks? Should some kind of functional data analysis including also the ridges themselves be attempted? And, if we use the current choice of landmarks, what is the effect of the analysis of using the particular polar co-ordinate scheme that was investigated by the authors? I very much hope that the authors will put all the data—both the particular *minutiae* and the full images—into the public domain in easily accessible form.

Another issue that is raised by the paper is the appropriate use and presentation of evidence of this kind in court. As I have indicated above, there is perhaps considerable work before a definitive data structure and model can be constructed for fingerprints, as opposed to the simple numerical structure of DNA profiles. In addition, I think that more effort needs to be invested in developing *mutual* understanding: in some instances statisticians and courts may be two communities divided by an apparently common language, and it behoves us as statisticians to try to obtain a clearer view of the legal reasoning behind court rulings on matters involving quantification of evidence.

One of the key recommendations of my recent report (Silverman, 2011) on research and development in forensic science, which was written in my role as Home Office Chief Scientific Adviser, was that Learned Societies should take an active role in promoting research in this area. I am therefore particularly pleased that the Royal Statistical Society has published this paper continuing its important contribution to forensic science research. But, finally, of course, I would like to add my thanks and congratulations to the authors of this most interesting paper.

Michael Risinger (*Seton Hall University School of Law, Newark*)

Although only a radical sceptic could reject the reliability of common source attribution when comparing two corresponding sets of rolled fingerprints (as in cadaver identification), the reasons for accepting the reliability of such attributions when one is dealing with a distorted and smudged crime scene fingermark of limited extent are neither obvious nor the subject of any critical scientific validity studies. What the authors have done in one masterful stroke of insight (followed by the requisite intense labour to instantiate it in a defensible empirical and statistical context) is to skirt the question of the validity of current practices, while laying the foundation for a different process that may someday displace current practices.

The key lay in a use of triangles that Pythagoras would have celebrated. They saw that the current system of describing a fingermark by using defined landmark points could form the foundation for specifying the corners of triangles, interlinked into polygons. They then took great care to study potential

distortion empirically, so that those triangles (and their relative connections) could be specified with a match window that is sufficiently large to overcapture a given triangle (or polygon) when a database of fingerprints is addressed. Such search thus gives an incidence for that fingermark's feature set within the database which is conservatively defined, so as not to overstate the rarity of the features in the fingermark. The more triangles the features of the fingermark allow you to generate, the more population incidence figures your database search will yield. Combine them and you can express the strength of the evidence as a well-warranted likelihood ratio.

This approach eliminates the current 'roll off the table' effect, where a fingermark is either attributed to the source absolutely or is classified as insufficient for use. The likelihood ratio statistic allows weaker but still warranted inferential weight to be assigned, and the likelihood ratios are well warranted because they rely on real data extracted from the fingermark and the database addressed. Finally, the approach provides one means of disciplining current practices, where some might be willing to push the conclusions that are warranted by a poor fingermark too far (though masking protocols remain necessary to control context bias).

It is unclear whether this approach can be applied to other phenomena, like handwriting, which lack the pre-existing *minutiae* specifications on which to anchor the triangles, and which have much larger intrasource variation than fingerprints: but, for fingerprints—brilliant!

Simon A. Cole (*University of California, Irvine*)

Not being a statistician, my comments will be, in turn, historical, sociological and philosophical.

First, readers may appreciate some sense of the historical significance of the paper. This paper represents a historic step in the development of a scientific foundation for making inferences about perceived associations between pairs of fingerprints. For historical reasons which brevity does not permit discussing, the taking of this first step is tardy; it arrives more than a century after the widespread implementation of fingerprint 'identification' as a practice and the acceptance of inferences about fingerprint associations as evidence in courts around the world, and it has occurred only after much heated denial of its necessity. Readers may be surprised to learn that as recently as the early 1990s it was argued that fingerprint evidence was non-probabilistic. Thus, the paper has a broader significance for the on-going, and somewhat begrudging, acceptance in law and forensic practice of the probabilistic nature of all evidence.

Second, I have a sociological comment. Forensic science has been faulted recently, with some justification, for having some characteristics of a 'guild' and maintaining insufficient ties to 'mainstream' science. Such claims necessarily implicate the credibility of specialty journals in forensic science to scrutinize adequately the research findings that they publish. The authors have set a commendable example for forensic science by submitting important research findings to the most rigorous scrutiny of the mainstream scientific community.

Last, I offer a philosophical–methodological comment. The goal of the model that is described is to allow a fact finder to estimate what weight to assign to a fingerprint association reported by a human fingerprint examiner. The model presented alone cannot answer that question without integrating information about the performance of human examiners. Perhaps the simplest way of articulating this point is to say that the model is *insensitive* to examiner performance: the findings reported in the paper would be the same regardless of the actual accuracy of practising examiners. Indeed, the very high likelihood ratios that are reported in this study are difficult to reconcile with what we know about actual casework errors and recent accuracy studies that suggest false positive error rates of at least 1 in 1000.

Likewise, performance data alone will not allow a complete estimate of the weight of a fingerprint association without the kind of rarity estimate that is made possible by this model. Both types of information would seem necessary.

David A. Stoney (*Stoney Forensic, Chantilly*)

This paper offers a practical validated method to measure the weight of fingerprint comparison evidence. By doing so it enables scientific analysis. We can describe, compare and systematically improve the process. We can test hypotheses and we can build a scientific foundation for evalution of fingerprint evidence. This foundation will ultimately replace one that relies on a mixture of traditional practices, untested assumptions and subjectivity.

One important consequence will be the abandonment of categorical expert judgements of fingerprint evidential value. These require an all-or-nothing approach: either a leap of faith that identifies *this* individual to the exclusion of all others or, failing in this, the affirmation that the fingerprint evidence has no value at all. For more than a century, this universal irrational practice has been in place as an unscientific compromise, for want of any alternative. This paper, at last, offers an alternative.

Another related, more subtle, consequence requires some introduction. The future of forensic science will focus more explicitly on the role of scientific evidence in legal decision making. The work of Biedermann *et al.* (2008) has *proven* the necessary and sufficient relationships connecting the *scientific finding* of the weight of evidence to the *decision process*, which is the *legal finding* of proof. There are three essential elements. The first is the weight of the evidence. Oversimplified, this is the frequency of occurrence of the evidence. Comprehensively and generally, it is the likelihood ratio, with relative support for alternative hypotheses.

The second element is the context into which the weight of the evidence must fit. This includes specific case circumstances, additional types of evidence and the priors that would transform a likelihood ratio into posterior odds.

The third element is the decision maker's own preferences: their assessment of risk and their thresholds for judgement. And it is only through the decision maker's judgement that the evidence becomes proof.

For over 100 years the courts and the public have expected, and fingerprint examiners have provided, expert testimony that fuses these three elements: offering testimony not as evidence, but as proof, assuming priors and including decision-making preferences. This created an overwhelming and unrealistic burden, asking fingerprint examiners, in the name of science, for something that science cannot provide. As a necessary consequence, fingerprint examiners became unscientific.

This paper provides a means to set things right: to cast off the unrealistic burden to provide proof, and to replace it with a validated offering of scientific evidence.

Ian Hunt (*London School of Economics and Political Science*)

The statistical methods in this paper are innovative and important. But I have two general comments.

(a) By quantifying evidence, statisticians justify both probabilistic *and categorical* beliefs.

Probability, interpreted as a partial degree of belief, can be justified statistically when calibrated with true frequencies or population proportions. Neumann and his colleagues advocate probabilistic beliefs and provide good candidate methods for calibration. But it is a mistake to write-off formal categorical belief, as Bayesian forensic statisticians frequently do. In fact, categorical belief dominates research in epistemology (Moser (2002), page 22). And legal scholar Ho Hock Lai (Lai, 2008) puts courtroom evidential reasoning on a firm categorical footing. It may be misleading to assert anything with absolute certainty, but it is not wrong either to believe something or not (with varying degrees and types of justification).

Statisticians are usually in court to give expert opinions or to offer facts that only experts can see properly (Malek (2009), chapter 33). Neumann and his colleagues take a Bayesian stance. They argue that statisticians should present probabilistic likelihood ratios to the court. So do Aitken *et al.* (2011). Sir David Cox and Deborah Mayo would probably dissent—their frequentist methods furnish quantitative support for various (refutable) *categorical* statements about hypotheses (Cox and Mayo (2010), pages 247–304). Like Bayesianism, frequentism entails probabilistic modelling. But modern frequentist results are expressed in terms of inferential error rates, confidence intervals and whether a hypothesis has 'passed a severe test'. Surely non-Bayesians can be experts also.

(b) Probabilistic modelling can be dangerous, disputable, or even irrelevant, for any *single case* before a court.

The foremost modelling concern for both Bayesians and frequentists is with the probability of a crime scene mark 'matching' an individual's police station print, conditional on the fact of the matter (the likelihood function, in other words). Neumann and his colleagues suggest innovative probability models, calibrated with representative databases of prints and marks. But, since fingerprint mark evidence is multifarious by nature and selection (surface difference, partialness, distortion, processing method, property definition, selection of *minutia* etc.), many different calibration options can exist for any single mark. This problem of categorization is known as the 'reference class problem' (Roberts, 2007; Hájek, 2007). It can cause probability conflicts, probability ranges and, in the case of no relevant or reliable classes, probability gaps.

Probabilistic modelling of fingerprint marks, with sufficiently broad reference classes, can help to identify which suspects to investigate fruitfully further; at this stage, modelling errors pose limited risk. But the stakes are high when an individual's case reaches the courtroom: justice may hang on reference class choices alone.

The following contributions were received in writing after the meeting.

Leonard G. Butt (*Scientific Working Group on Friction Ridge Analysis, Study and Technology, Pikesville*)
My comments are prepared purely from the perspective of the practical application of fingerprint identification by a forensic scientist.

For more than 100 years the use of fingerprints as a means of personal identification has been widely applied and accepted throughout the world. The weight that has been placed on such work by many within our society cannot be overstated. Unequivocally the most significant of such would be instances involving the identification of marks that have been recovered from crime scenes which led to a conviction and the death penalty.

With very limited exception, the measurement by which fingerprint identifications are confirmed has always been subjective. The fingerprint community would like to be able to state that, despite the subjectivity, the results reported are always certain and dependable; however, the fact is that in an unknown number of cases there have been errors. There are many factors which can be pointed to as causes for such instances to have occurred. What is important, however, is that errors do exist. Furthermore, the end user of the services that are provided by fingerprint examiners are not and should not be in the position of questioning every result that is provided to them.

Many of us in the field are excited about the potential of having access to a tool which is capable of evaluating the information used by an examiner in reaching their decision and providing an objective assessment of the probability that two impressions originated from the same source. It is time to move forwards; to provide another level of objective measurement to support the good work of many and to reveal the poor work of others.

It is sad to consider that too many examiners appear to lose sight of the fact that the decisions they render can have a monumental effect on an individual. Rather, they become engulfed in the personal challenge to identify a print in a particularly difficult case. Such conditions generate the perfect opportunity for the failure of a subjective system.

From the courts, to the prosecution, to the defence, to the victims and to the examiners themselves the importance of an effective application such as that presented in this paper cannot be overstated. I and the entire community anxiously await the results of the vigorous assessment that the Royal Statistical Society will give this paper.

José E. Chacón (*Universidad de Extremadura, Badajoz*)
I congratulate the authors on making a considerable and successful effort to tackle such a difficult problem as quantifying evidence weights from fingerprints.

In Section 8, the authors circumvent the problem of modelling the probability density function π in the likelihood ratio by introducing a weighting function ψ on the distances between the observed data and the simulated marks. This can be regarded as a plausible non-parametric alternative to the problem of model specification in likelihood ratio tests but, as the authors themselves recognize, the choice of the weighting function is somewhat arbitrary and open to criticisms.

Nevertheless, it should be pointed out that there are statistical models on *minutiae* features that have been proposed recently in the literature. For instance, Zhu *et al.* (2007) suggested a mixture model on the location and direction of *minutiae* where each component in the mixture has the location modelled according to a bivariate normal distribution and the direction modelled as a von Mises distribution, and derived an EM algorithm to approximate the maximum likelihood estimation of the parameters, which can be used as a starting point to develop a likelihood ratio statistic to measure the degree of coincidence between a mark and a control print in a parametric setting.

But still in the non-parametric context Pardo *et al.* (2004) developed a partition-based generalized likelihood ratio test of homogeneity which could be useful as well to compare the distribution of the *minutiae* features of the mark and the print. A further non-parametric approach, although not based on likelihood ratios but on the L_1-distance, was presented in Biau and Györfi (2005) for testing whether two samples are homogeneous. An empirical comparison of all these procedures with that suggested in the present paper would be of great interest.

Nick Fieller (*University of Sheffield*)
Along with other contributors I enjoyed both reading the authors' paper and hearing it presented and I welcome the advances in statistical methodology presented. The discussion at the meeting raised many issues and doubtlessly many more will be raised in the written contributions. The key statistic that is provided by the authors' calculations is a likelihood ratio and a particular concern mentioned by some discussants was the issue of use of this in a Bayesian assessment of the evidence presented. Although this

may be a natural approach for some people there is no intrinsic reason for doing so. The likelihood ratio can be used and interpreted directly as a measure of the weight of the evidence of the prosecutor's case in relation to the defendant's. Naively, if the ratio is greater than 1.0 the evidence favours the prosecution case; less than 1.0 the defence case. However, in practice some calibration of the measure is required. This can be achieved by examining its operating characteristics on known matches and non-matches (as the authors reported). Almost certainly, there needs to be an indeterminate band where the evidence must be regarded as inconclusive. In the case of glass fragments that was reported by Aitken and Lucy (2004) I gather that they felt that a band of about 0.3–3.0 was a reasonable grey area to avoid to prevent making too many false positive or negative decisions (Lucy, personal communication). In other contexts the grey area needs to be substantially wider.

In work that colleagues and I did on facial comparisons recently (reported in Evison and Vordebrugge (2010)) we found from comparing good quality photographs of known non-identical subjects that a grey band from 0.003 to 300.0 was required to keep the rate of false decisions to reasonable levels. We interpreted the wider band as reflecting the considerable uncertainty in manual placement of landmarks on facial features (rather more than 30 landmarks were used) and in subtle changes in expression in faces. In the current paper the authors ingeniously model the analogous distortions in fingermarks under varying degrees of pressure but nevertheless I suspect that a range of 'no decision possible' is required rather than a strict switch in decision at the borderline of the likelihood ratio equal to l. Perhaps the authors could comment on whether they would recommend such an indeterminate area.

Stephen E. Fienberg (*Carnegie Mellon University, Pittsburgh*)
I congratulate the authors on an excellent and convincing paper. It represents a major advance at the interface of statistics and forensic science. They make clear the relevance of the methodology for the task of presenting fingerprint comparison evidence in a legal setting, and they have designed and implemented a painstaking empirical evaluation that clearly informs the quantification of such evidence. The empirical evidence in Fig. 3 is especially compelling in resolving the controversial issue regarding the inappropriateness of a numerical points standard. I have two questions. First, how do you square the evaluation in your paper with the high profile instances that we have seen in the USA of fingerprint comparison errors, e.g. in matching partials to the automated fingerprint identification system database as in the Madrid bombings and Mayfield case in 2004, and in the experimental evidence of errors in studies such as by Ulery *et al.* (2010)? Second, much has been made about technical advances in fingerprinting such as the use of wavelets in the automated fingerprint identification system and Gabor filters (Jain *et al.*, 2000) and how they achieve greater than 99% accuracy in tests done by the National Institute of Standards and Technology. If I read the paper correctly, this kind of technology has not yet been able to overcome the fundamental incompleteness of partial print evidence left at the scene of a crime, and, since this is a primary source of fingerprint comparison error, it does not promise to increase the accuracy of fingerprint evidence in the near future. Is that correct?

J. L. Gastwirth and Q. Pan (*George Washington University, Washington DC*)
The paper proposes a novel approach for extracting information from fingerprint marks to measure the degree of agreement between two. Several interesting issues deserve further study.

(a) The authors demonstrate that the procedure provides well-separated log-likelihood ratios from matches and non-matches. However, the ultimate goal is to provide a conclusion to the court. Recommendations on the selection of cut-off and the type I (reports a non-match as a match) and type II (misses a real match) errors would be helpful. Finally, courts should be informed about the degree of uncertainty (probabilities of type I and type II errors) accompanying a claimed match or mismatch.
(b) Each *minutia* is described by a five-element vector and the similarity of two *minutiae* is a weighted average of elementwise differences. The weights were chosen to minimize the false acceptance and false rejection rates on the basis of a reference database. Alternative dimension reduction methods, principal component analysis, logistic regression or sliced inverse regression, provide other ways to summarize high dimensional data ($\Delta(l, k)$) to differentiate between match and non-match.
(c) The choice of *minutiae* was done by trained experts because human brains recognize informative patterns better than computers. Experts, however, are subject to bias (Dror *et al.*, 2006) especially when they are aware of information from the criminal investigation. How can the method be implemented to minimize subjective influence? Can it be standardized and automated?

(d) The analysis assumes (assumption 5) that

'the k-configuration in the print that is selected by the examiner as closest to the mark is the configuration that has minimum distance from the mark'.

Humans are not perfect; what is the effect on the likelihood ratio if the second-closest configuration were selected?

(e) The authors use the likelihood ratio based on a fairly complex linear combination of distances and one wonders how easily it will be to communicate their results in a manner that is accessible to jurors. Perhaps Chernoff (1973) faces using the data extracted from several *minutiae* configurations to determine the features of the face would provide an intuitive way of demonstrating that the fingerprints at the crime scene match those of the suspect much more closely than a randomly chosen set from a database.

(f) Computational concerns limited the number of *minutiae* to no more than 12. The log-likelihood ratio in Fig. 5 under H_p plateaus at about 10 *minutiae*, suggesting that little additional power is obtained from more *minutiae*.

Ramón Giraldo and Carlos Alonso Malaver (*Universidad Nacional de Colombia, Bogata*) **and Emilio Porcu** (*Universidad de Castilla la Mancha*)

We congratulate the authors for this beautiful paper, for the excellent historical introduction to the subject and for the important evaluation of the model that they propose, which is the same model as in Neumann *et al.* (2007). We have some comments that may be considered by the authors.

(a) In Section 1 the authors mention limits and drawbacks of earlier methods, with particular emphasis on the limiting assumption of independence between the characteristics of the fingerprints and, most importantly, of spatial independence between them. Despite this, this aspect is almost not considered in the paper. Several researchers have gone into the direction of spatial dependence, and in particular spatial point processes have proven to be very useful for this purpose (see Lim and Dass (2011), and the references therein).

(b) The model proposed is tested with several data sets. Some of these data dispose of the Cartesian co-ordinates of particular features such as cores and deltas in the ridge flow. Such information could be used to implement the spatial analysis that was mentioned in point (a).

(c) A more technical comment derives from the likelihood ratio as in equation (6), for which we have that it depends on the sample size N (in the denominator), which makes it necessary to study its sensitivity with respect to N when it becomes large.

(d) In the model on page 378 (the number of *minutiae*), one may want to consider k as a random variable, which implies redefining accordingly the likelihood ratio and obtaining more general and realistic models.

Thomas Hotz and Axel Munk (*Georgia Augusta University of Göttingen*)

We praise C. Neumann, I. W. Evett and J. Skerrett for raising before the statistics community the important question about the weight of evidence that a fingerprint comparison carries, initiating this essential discussion. Indeed, attributing a statistical measure of certainty to the evidence presented appears indispensable. The likelihood-based approach proposed is a major step in that direction. We believe that further advances can be achieved by taking the following points into account.

(a) Apart from the concrete model being used, we are somewhat worried that assumption 1 may not hold in practice: the innocent suspect may conceivably have been accused *because* his control X appears similar to the trace Y. This is most obvious when the suspect has been determined as the closest match from a database search, *biasing* X to look more like Y than a control independently drawn from the population. In fact, the authors took this into account when validating their method: they 'retrieved' 'the most similar control print in the database' (Section 5.5). From a statistics point of view, this raises issues in multiple testing which need to be addressed.

(b) The authors compare triangles by using several features (radius, side length and area (Table 1)), finding that area did not contribute much after taking the others into account. However, there is a well-developed theory within shape analysis (Dryden and Mardia, 1998) allowing us to compare triangles and model distributions of these; see Gottschlich *et al.* (2011) for an application of shape analysis to fingerprints. In fact, entire polygons comprising ordered vertices, or being even star shaped as considered here, may be represented in a Euclidean space, rendering their analysis

particularly simple (Hobolth *et al*., 2002; Hotz *et al*., 2010), as all methods of multivariate analysis (Mardia *et al*., 1979; Anderson, 2003) become directly applicable. Such models will intrinsically respect correlations of the individual variables, e.g. caused by the closedness condition of the polygon's boundary, though they would need to be extended to include the directions of the *minutiae*.

(c) The proposed negative log-likelihood is proportional to the weighted sum of squared distances between the features (equation (7)), the distances being arbitrarily chosen. Again, multivariate analysis, e.g. using principal components, could take correlations between the features into account, the aim being to come up with a likelihood which is less model based and thus subjective but has rather been learned from training data. This idea could be carried further by using multivariate non-parametric density estimators; see for example Wand and Jones (1995).

Venkata K. Jandhyala and Stergios B. Fotopoulos (*Washington State University, Pullman*)

Categorical statements by forensic experts regarding fingerprint comparisons are solely dependent on the 'expert' and are not open for review. In contrast, quantitative evaluation of a potential match based on a widely acceptable model makes immense sense, and it is in this regard that the present paper introduces a paradigm shift.

The central issue is whether the model developed in the paper is in a form that is widely acceptable to the legal and forensic community. As the authors themselves note, the model development in the paper is open for review and further improvements. Among the many issues that may be raised, we point out the following. First, it is wholly based on identification of *minutiae*. Secondly, although the assumptions made are seemingly convincing, some of them should be put to rigorous testing and evaluation. Thirdly, the method of arriving at the weight function (14) is mostly data specific and thus awaits further validation and scrutiny. For brevity, we limit our further comments to the first issue only.

First the term *minutia* was not defined in the paper. Since the paper is predominantly aimed at statisticians, the authors could help the reader by introducing *minutia* as a location where a ridge ends or bifurcates. Whereas the authors focus on capturing reproducibility of labelling *minutiae*, they seem to have ignored proper identification of *minutiae* themselves. Also, one has the impression that data on fingerprints are trustworthy, and marks alone may have distortions. However, fingerprints themselves are subject to extraction errors (see Maltoni *et al.* (2009), chapter 4) such as overlaps, non-linear distortions and skin condition. Thus, importance should be given to proper identification of *minutiae* themselves in developing the model.

Finally, Srihari and Su (2011) recently proposed probability of random correspondence as a way of quantifying fingerprint matching. Their model development is based on capturing the distribution of *minutiae* locations and their orientations. Proposing a Gaussian mixture for *minutiae* locations and von Mises distribution for orientations, they pursued parameter estimation on the basis of the EM algorithm. They found good results when they implemented their model to fingerprint data from the National Institute of Standards and Technology.

Although the model developed by Srihari and Su (2011) appears to be easier to implement, its performance on large data sets has not yet been put to test. On balance, comparison of such competing models for quantification of the weight of evidence goes a long way in the process of refining the methodology proposed.

Joseph B. Kadane (*Carnegie Mellon University, Pittsburgh*)

The progress that is reflected in this paper and its predecessor (Neumann *et al.*, 2007) to inference about fingerprinting is indeed to be celebrated. However, further work is needed.

This paper gives a careful argument leading to the likelihood ratio (6), which is further specified in equation (9). However, the argument from equation (9) to equation (14) is basically, like that in Neumann *et al.* (2007), heuristic. It comes down to having tried a few different specifications and choosing one that has operating characteristics that are believed to be favourable. But why should one's beliefs be captured by equation (14)? And, without such justification, what warrants calling the result a likelihood ratio?

If the point were simply to find a rule that correctly classifies the sample cases, why not use a machine learning automatic classifier, unrestrained by reference to likelihood ratios? This path involves danger of overfitting. I favour the approach leading to equation (9), but I think that whatever specification is used for the function π must be explained probabilistically. If the result has good operating characteristics, well and good. If it does not, perhaps this finding would suggest that fingerprint evidence is not as strong as we have been led to believe.

David H. Kaye (*Pennsylvania State University, University Park*)

Will the courts and latent print examiners embrace likelihood ratios?

Neumann, Evett and Skerrett have devised an impressive automated procedure for taking *minutiae* as discerned by fingerprint examiners on crime scene marks, deriving a score based on relative positions and other data, and generating a likelihood ratio (LR) for the hypothesis of a common *versus* a disparate source. They propose further refinement, testing and then widespread application to criminal cases. They suggest that the statistical procedure ultimately can overturn the dominant paradigm of categorical testimony of absolutely certainty.

Such a shift should be welcome in Anglo-American legal systems. Prominent judges have called for more qualified conclusions than the usual assertions that a mark could not possibly have come from anyone else on earth (Edwards (2009) and Leveson (2010), page 14). But there are issues in ascertaining and communicating the uncertainty in LRs (Morrison, 2011), and it takes discipline to present 'likelihoods' and LRs so that they will not be misconstrued as posterior probabilities or odds (Kaye *et al*., 2011). Furthermore, English courts have expressed misgivings about likelihood ratios (especially when based on subjective probabilities, as in R. *versus* T., 2010, EWCA Crim 2439). Nevertheless, there is ample precedent for their use, primarily from DNA cases with mixed samples and from child support, immigration and criminal cases that rely on kinship testing (Kaye, 2010; Kaye *et al*., 2011), and describing a suitably computed LR is a reasonable way to assist legal factfinders to draw their own inferences (Kaye, 1982).

Before numerical LRs enter the courtroom, however, the scientific validity of the particular procedure for computing them must be established by showing sufficient testing and published validation research (in most US jurisdictions) or by showing general acceptance in the scientific community (in many other states). There are no hard-and-fast rules about how much research is enough. The very publication of the study in this journal (without ensuing controversy in the statistical community) will go a long way towards meeting these standards.

In the past, latent fingerprint examiners have not been well educated about statistics and have been willing to present misconceived statistical analyses in court (see, for example, Kaye (2003)). Will they quickly embrace the brave new world of LRs? Or will Planck's (1949) view (pages 33–34) that

> 'a new scientific truth does not triumph by convincing its opponents and making them see the light, but rather because its opponents eventually die, and a new generation grows up'

prevail?

Steffen Lauritzen (*University of Oxford*), **Robert Cowell** (*Cass Business School, London*) **and Therese Graversen** (*University of Oxford*)

Firstly, we congratulate the authors for tackling this important problem and providing a plausible approach for quantification of fingerprint evidence.

We understand that $\pi(y|x)$ denotes a generic distribution of features of a mark y taken from a source x, which in itself is independent of the various hypotheses so we have

$$f(y \mid H_{\mathrm{d}}) = \frac{1}{N} \sum_{i=1}^{N} \pi(y \mid z_i),$$

and

$$f(y \mid H_{\mathrm{p}}) = \pi(y \mid x),$$

leading to the likelihood ratio

$$\mathrm{LR} = \pi(y \mid x) \Big/ \sum_i \pi(y \mid z_i).$$

In the paper this model is further developed as eventually implying

$$\pi(y \mid x, q, \lambda_1, \lambda_2) = \psi\{d_q(y, x), \lambda_1, \lambda_2\},$$

where d_q is a distance measure and λ_i are unknown parameters. The parameters $(q, \lambda_1, \lambda_2)$ are then optimized by considering the discriminatory power of the associated procedure. However, if the likelihood ratio is interpretable as such, it seems to make more sense to optimize these parameters simply by generic estimation methods, such as maximum likelihood. Also, it would be more convincing if the fit of this distribution $\pi(y \mid x)$ were properly investigated. If this fit was very bad, the procedure described seems to

be no more than an *ad hoc* rule for quantifying evidence and the term likelihood being used somewhat out of place.

We find that the distance measure that is described in formula (7) is somewhat unconvincing and would rather prefer the model $\pi(y \mid x)$ based on a generic distribution of feature selection and distortion; see for example what is discussed in the literature of *Procrustes problems* and *alignment* in for example bioinformatics (Green and Mardia, 2006; Dryden and Mardia, 1998; Gower and Dijksterhuis, 2004). The alignment problems seem to us to have much in common with the problem that is discussed here.

Finally, as also prints are prone to distortion error, we would rather calculate the likelihood ratio from assuming a true feature vector x, a print feature x^* and a mark y and then considering $f(y \mid x^*)$ where in general we would have $f(x, y, x^*) = f(x^* \mid x, y)\, f(y \mid x)\, f(x)$, reflecting that the selection of features in x^* can depend on the mark y.

Stephanie Llewelyn (*University of Sheffield*)

I congratulate the authors on their thorough and well-thought-out paper. The approach based on likelihood ratios is widely applicable and is the core to the work I have been doing sponsored by the Forensic Science Service and the Engineering and Physical Sciences Research Council. Whereas they have relied on empirical densities I have looked at parametric spatial models for the pattern of the two main types of *minutiae*: ridge endings and bifurcations. In passing, I might comment that I have found little evidence of substantial differences between fingers in statistical properties of the point patterns of *minutiae*, which was a question that was raised in the discussion. The approach that I have used does rely on complete observation of all points within a moderate region. The parameter estimation would be disrupted if one or more points in a fingermark inside a region were not observed but were present in the corresponding fingerprint on the database, Do the authors have any information on how robust their procedure is to similar problems? This leads me to the question of how the k-configuration is chosen in each fingerprint so that the configuration from the mark can be searched against these. If a typical fingerprint has 120 *minutiae*, for example, do the authors search every possible 10-point configuration from the 120 *minutiae* with every print in the database to find the configuration with the smallest distance? If not, how do they restrict the number of searches to be made? Is it two configurations which have no other *minutiae* interior to the polygon that is formed by the configuration or is this done by an examiner who determines a likely subset from the whole fingerprint?

Inevitably there are many loose ends in work such as this but one in particular is of key interest. This is the question of several fingermarks at a crime scene which an expert 'knows' to be from the same hand or person (e.g. on a glass). How would the authors propose to combine the evidence obtained from the separate marks?

Jorge Mateu (*University Jaume I, Castellón*)

The authors are to be congratulated on a valuable contribution and thought-provoking paper in the context of forensic identification. A contribution emphasizing the role of statistical modelling in helping with real social problems, such as evaluation of evidence in fingerprint comparison, is greatly welcome.

I would like to focus on the definition and the building blocks of the likelihood ratio LR, assuming that we have a trace, a control and a set of database members. This LR depends on the way that we compare the relative positions of a set of pairs of *minutiae* between a mark and a print in two distinct configurations. Note that each configuration is defined through k *minutiae*, where each *minutia* is formed by five features, variables measured on a print or mark. So we are basically comparing a set of two k-configurations of five features each. The authors develop in Section 8 a way to handle LR through functions of particular distances, ignoring possible realistic spatial dependences between the variables defining the features within a *minutia,* and between *minutiae*. This is my point here. One possible and realistic (although perhaps more mathematically tricky) approach to build the ratio LR is to consider the 5-size vector of features as a random vector exhibiting spatial dependences between the features within a *minutia*, and possible spatial dependences between the *minutiae*. These dependences should be incorporated in the definition of the distances in equations (7) and (10), and in the definition of the weighting function in equation (10). The expressions (11)–(14) for the weighting function could be extended to conform to this more general framework. To have some food for thought, and thinking out loud, here are some questions of interest:

(a) which might be the adequate statistical method that can compare two k-sets of 5-size random vectors containing spatially dependent components;

(b) assuming that we have a mark and a print coming from related people, can we modify the distance function to handle clear links between the several *minutiae*?

A related note comes from asumption 1. As the authors themselves state, there are situations where this assumption is invalid. Consideration of relatedness issues in fingerprint comparisons could come from considering spatial issues in the components of the weighting function.

Robert S. Ramotowski (*Arlington*)
The importance of visualizing and comparing latent friction ridge detail to effect an identification of potential suspects cannot be overstated. Interpol recognized the importance of such evidence in a 2004 report, noting that

> 'In the investigation of crime, fingerprints found on the scenes lead to more suspects and generate more evidence in court than all other forensic techniques combined'

(Interpol European Expert Group on Fingerprint Identification, 2004).

In the USA, the admissibility of fingerprint evidence began to be challenged in the late 1990s. In U.S. *versus* Mitchell, the defence raised the issue of whether or not there is

> '... a scientific basis for a fingerprint examiner to make an identification, of absolute certainty, from a small distorted latent fingerprint fragment, revealing only a small number of basic ridge characteristics...'

(145 F.3d 572 (3rd Cir 1998)). In 2009, the National Academy of Sciences released a report titled 'Strengthening forensic science in the United States: a path forward,' which highlighted the use of subjective criteria by fingerprint examiners to reach a conclusion of individualization. The committee recommended that

> '... more research is needed regarding the discriminating value of the various ridge formations and clusters of ridge formations'

(National Research Council, 2009).

Although there have been attempts to establish the probability of finding similar clusters of friction ridge *minutiae* dating back to Sir Francis Galton's study in 1892, no single effort has been widely accepted by either the scientific or judicial communities. However, the recent model developed by Neumann and his colleagues in this paper has begun to establish itself and to gain acceptance within the fingerprint community. Although not intended to replace the examiner, the likelihood ratios that are generated from this statistical model have considerable value as both supporting evidence for fingerprint identifications and a quality assurance tool to measure the risk of associating close non-matches.

The influence of this new statistical model on the fingerprint community was recently demonstrated. On July 16th, 2010, members of the International Association for Identification passed resolution 2010-18, which rescinded the restrictions that had been imposed in 1979 (resolution 1979-7) on examiners for providing '... any testimony or reporting of possible, probable, or likely friction ridge identification[s]'. The new resolution stated that

> 'The use of mathematically based models to assess the associative value of the evidence may provide a scientifically sound basis for supporting the examiner's opinion'.

Although not yet fully validated, the model that was developed by Neumann and his colleagues played a role in overturning this three-decade-long ban on the use of statistical probabilities in fingerprint comparisons.

Bernard Robertson (*New Zealand Law Journal*), **G. A. Vignaux** (*Victoria University of Wellington*) **and Charles E. H. Berger** (*Netherlands Forensic Institute, The Hague*)
We congratulate the authors on this application of the logical approach in comparing forensic fingerprints. In 1995 two of us tried to make clear that logic applies to any forensic area of expertise (Robertson and Vignaux, 1995). Given any particular level of knowledge and understanding, however, it may be more difficult to apply in some areas, such as fingerprints, than in others. Some have interpreted this concession as making an exception for fingerprints, and have suggested extending the same exception to other areas, such as tool marks and shoe marks (Rudram, 1996; Aitken and Taroni, 1996, 1997). Such an exception was not intended, is undesirable and, as the present paper clearly demonstrates, should not be necessary.

Several forensic disciplines have since developed objective methods of interpretation based on probability. The International Association for Identification has rescinded the resolution that forbade members

to give testimony in terms of probability and has now resolved that

> 'The use of mathematically based models to assess the associative value of the evidence may provide a scientifically sound basis for supporting the examiner's opinion'

(resolution 2010–18). The Board of the European Network of Forensic Science Institutes—an organization with 58 member laboratories in 33 countries—has engaged itself to implement the logical approach in its member laboratories. Clearly, though, more work is still to be done.

The present paper is an important step towards determining the value of fingerprint evidence.

We would like to ask the following questions.

(a) The value of the evidence depends crucially on the within-source variability, and the authors attempt to model variability due to distortion and *minutiae* assignment. However, the performance of the model is discussed qualitatively in terms of misleading evidence. Is it possible that the same performance would have been obtained without taking these sources of variability into account?
(b) How do the authors view the magnitude of the likelihood ratios considering the spread in their values, and the nine out of 122 instances of misleading evidence in the 'small experiment'?
(c) Could the authors compare their work with that of Gonzalez-Rodriguez *et al.* (2005)?
(d) Why did the authors not use one of the existing fast computer algorithms that can produce thousands of univariate comparison scores per second in favour of their own comparison algorithm?

Michael J. Saks and Ashley M. Votruba (*Arizona State University, Tempe*)

Significance

Neumann, Evett and Skerrett have made a major contribution to the art and science of fingerprint identification. This is an important—perhaps historic—-step forward in the intellectual history of fingerprint identification and perhaps other fields of pattern matching forensic science.

Their work deals ingeniously with the elusive problem of placing forensic identification on an empirically sound, quantitative foundation. It paves the way for moving forensic identification away from a century of overwhelmingly subjective judgement and categorical conclusions. (The absolutist nature of fingerprint expert opinions is largely attributable to the imperatives of legal advocacy overpowering the imperatives of science.) It also directly confronts complexities in fingerprint observation and measurement that are glossed over in conventional forensic analysis: intraprint and intramark variation, variation between examiners, difficulty of harnessing available databases, quantification of the weight of the evidence, and so on. Other laudable aspects of the work include making explicit the authors' assumptions and acknowledging the complexity of the problems of *scientific* forensic identification.

Implications for the future

Most obviously, the approach of Neumann and his colleagues allows the fiction of absolute identification to give way and the results of fingerprint examinations to be expressed in probabilistic terms. The precise expression remains to be worked out, and the implications (if any) for the trial process need to unfold. Examiners who previously refrained from testifying because they felt less than 100% certain of identification can testify more often. Jurors might perceive reasonable doubt despite a fingerprint identification. The question arises, then, about how probabilistic forensic identification evidence will interact with other evidence in a case.

The new paradigm calls on us to consider potential difficulties surrounding implementation. Suppose that a cost–benefit analysis compared the computer-aided, quantified approach with the conventional approach and found higher costs but otherwise no differences. Is the justice system willing to pay more for something solely because it more closely approximates truth?

Even if the method of Neumann and his colleagues did not become the approach of choice in routine casework, it could nevertheless be used in research that would help to set lower and upper bounds on the weight of evidence, given certain facts about the print and mark under examination. That would permit experts to testify to weight-of-evidence within certain calculated bounds.

The ultimate destination of such research on forensic identification is full computer automation, free of idiosyncratic human judgement, and potentially offering full transparency—though susceptible to its own limitations and imperfections.

Hal Stern (*University of California, Irvine*)

Neumann, Evett and Skerett have provided a major advance in the quantitative assessment of fingerprint evidence. The ability to obtain a numerical measure of the weight of the evidence that is provided by a

mark and associated print will do much to address the loss of information that results from the traditional use of identification, exclusion or inconclusive as the only permissible comparison outcomes. The work is especially exciting because the authors have provided a comprehensive treatment of many sources of variability that are intrinsic to the analysis of latent prints. There are two points that I would like to make in this brief comment. First, the continued use of the term 'likelihood ratio' after the authors' transition from equation (9) (where π is a probability density function) to equation (10) (where ψ is a weighting function) may be misleading to users. The motivation for the transition is clear; the probability density function is of very high dimension ($4k$ or $5k$ for k *minutiae*) and neither a parametric nor a non-parametric density estimate seems practical for evaluation of equation (9). The likelihood ratio has a well-established statistical meaning; for one example in the standard Bayesian analysis of forensic evidence the likelihood ratio is the multiplier that converts prior odds into posterior odds. If the authors' quantity LR is not a likelihood ratio then it is probably better to avoid the term. The numerical results in Section 9 of the paper convince me that LR is of value to the forensic world regardless of whether it is a formal likelihood ratio or not. A second point is that the paper does not address the robustness and sensitivity of the authors' approach to the various choices that have been made in developing the approach. For example, the authors have optimized their distance metric, determining which features to include and how to weigh them. How much do results vary when other choices are used? In the same vein, how sensitive are the LR results to the parameters that characterize the examiner influence model or the fingerprint distortion model? Providing results of this type will be crucial for gaining broad acceptance of the approach, especially given the diverse backgrounds and interests of the many stakeholders. I look forward to continued development and empirical investigation of this innovative approach to assessing fingerprint evidence.

William C. Thompson (*University of California, Irvine*)

This ground breaking and otherwise excellent paper has an important weakness: it fails to consider that observer effects and confirmatory bias may influence latent print examinations. As currently implemented in most laboratories, the analysis–comparison–evaluation–verification protocol allows an examiner's interpretation of a mark to be influenced by details of the print, and vice versa. Particularly when there is reason to believe that the mark and print have a common source, examiners may tend to disregard or explain away inconsistencies, while interpreting ambiguous data as confirmation of a match, as did the Federal Bureau of Investigation examiners in the notorious Mayfield case.

This interpretive bias may cause examiners to see more similarity in prints from different fingers than did participants in the reported studies (who evaluated marks without knowledge of the comparison prints, and vice versa). Consequently, in actual practice the probative value of an examiner's subjective determination that a mark and print have a common source may be less, and perhaps considerably less, than the reported validation studies suggest. Although this paper is an important step in the right direction, the claim that examiners can determine with near certainty that a mark was made by a particular finger requires more validation under more realistic conditions.

The potential for confirmatory bias also undermines the validity of the likelihood ratios that are derived from the quantitative model proposed. The model assumes that *minutiae* are detected and labelled independently in the print and mark. Whereas that assumption might be valid for examiners who follow procedures designed to minimize observer effects (e.g. 'linear analysis–comparison–evaluation–verification' or 'sequential unmasking'), it is not valid for most latent print examiners because such procedures have yet to be widely adopted.

The problem is not simply variability between examiners in their scoring of *minutiae* but a tendency for examiners to 'paint the target around the arrow' by allowing their interpretation of the print to influence their interpretation of the mark. To the extent that this confirmatory bias occurs, the likelihood ratios generated by the model will overstate the value of examiners' individualizations. Hence, latent print examiners should be cautious about incorporating likelihood ratios derived from the proposed model into their testimony and reports. If these statistics are presented at all, it should only be in cases in which the latent print examination was conducted under protocols designed to minimize observer effects and confirmatory bias.

Max Welling (*University of California, Irvine*)

Let me begin with congratulating the authors on their landmark paper. This is perhaps the most comprehensive study to date on statistical fingerprint analysis.

My main critique concerns whether these likelihood ratios LR can be considered realistic. In other words, can we really interpret $\mathrm{LR} = 1$ (or $\log_{10}(\mathrm{LR}) = 0$) as evidence neither in favour nor against a true match? In Fig. 6 I have plotted a histogram of the $\log_{10}(\mathrm{LR})$-values (scaled so that the highest peak equals 1).

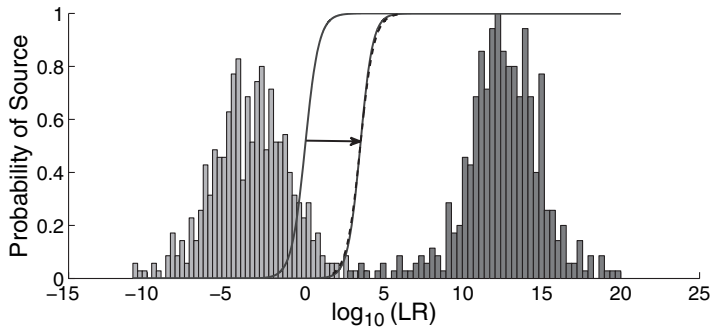


Fig. 6. Histograms of the log(LR)-values (scaled so that the highest peak equals 1) for non-sources (□) and sources (■): the left-hand rising curve represents the empirical probability of the same source (true match), P(source), whereas the shifted curve to the right of it represents the recalibrated probability of the same source; the large discrepancy between these curves implies that the empirical LR-values are biased three orders of magnitude in favour of a true match

Dark bars count non-sources and lighter bars count sources. We observe a nice separation in LR-values between sources and non-sources. The rising curve on the left is the probability of same source (true match), P(source), computed from these LRs, assuming equal prior probabilities. The empirical P(source) is the curve on the right computed from performing a logistic regression on the LRs (I thank the authors for sharing their data). We can make the curves overlie by assuming P(non-source) $= 3120\ P$ (source), as is also shown in Fig. 6 (owing to close match the two curves are difficult to distinguish).

This result implies a bias of more than three orders of magnitude in favour of interpreting an LR as a true match! In other words LR $= 1$ really means that it is 3120 times more likely that there is no true match whereas LR $= 3120$ corresponds to evidence in neither direction.

My observation is not necessarily an objection against the model but against interpreting the uncalibrated LRs as realistic evidence. All models are wrong and should be recalibrated empirically. In this case, we observe that no correlations between the *minutiae* attributes (Table 1) are modelled, resulting in an overcounting of evidence (in either direction). Moreover, the method involves 'look[ing] for the k-configuration in each of the prints that minimizes the distance from the mark'. Selecting the k best matching *minutiae* will cause prints with a higher number of identified *minutiae* in the database to score higher than prints with a smaller number of identified *minutiae*. The correct, more computationally expensive procedure involves a noise model for falsely detected *minutiae* and 'marginalizing' (rather than minimizing) out unobserved minutiae. Also, the model that is entertained in equation (14) is unlikely to describe nature perfectly and so on.

In conclusion, without recalibration, the LR-values that are computed by this model should not be used in court. The good news is that recalibrating the model is straightforward but this final analysis seems missing from the paper.

The **authors** replied later, in writing, as follows.

As we explained in Section 1, our work is aimed at enabling the quantitative evaluation of evidence that is currently reported as categorical opinions. The model that we developed and evaluated is similar in principle to those proposed for other types of evidence, particularly glass (Lindley, 1977) and DNA evidence (Evett and Weir, 1998). However, the essence of our problem is different from these in the sense that there is no simply identifiable set of variables to describe a fingerprint pattern, as opposed to refractive index for glass or allele designation for DNA. The implications are numerous and the points made by the discussants can be summarized in four broad categories:

(a) the statistic resulting from our model is not formally a likelihood ratio;
(b) the choice of the variables that are used to describe fingerprint patterns in our research may not be optimal, nor capture the entirety of the information on those patterns;
(c) the outcome of our model can be influenced by users, and by assumptions made regarding the reference database;
(d) ultimately, the weight of evidence that is assigned to a particular fingerprint comparison will need to be reported in court and the process by which the weight was assigned may need to be explained to judges and members of a jury.

Several discussants make powerful and valuable suggestions with regard to modelling. We do not wish to give the impression that we consider our model as the last word: on the contrary we recognize the scope for much improvement. We think that the point made by Kadane and Stern that what we calculate is not actually a likelihood ratio tends towards the academic. It seems to us that, trivial situations apart, the likelihood ratio will always be model dependent. In the forensic *milieu* it becomes part of a grossly imprecise process. There is no need for the unattainable ideal of a likelihood ratio that is precise to four significant figures—a fair and reasonable assessment of the order of magnitude clearly works with DNA evidence. We seek no more.

Our model is different from most models proposed recently (Dass *et al*., 2009; Su and Srihari, 2009). Indeed, it does not attempt to model the wealth of information that is theoretically present on fingerprint patterns to give some general measure of their rarity and it does not rely on global assumptions regarding the dependence between the features on these patterns. We chose to focus on the quantification of the weight of fingerprint evidence in individual cases by considering the features that are observed on marks and their spatial dependence. As noted by Stern, we conveniently chose to reduce the dimensionality of the problem by computing distances between vectors of variables instead of modelling the distributions of these variables. Gastwirth and Pan commented that we could use other data reduction methods. In this project, we followed the path of the developers of automatic fingerprint identification system algorithms, who have demonstrated that a one-dimensional variable can efficiently capture the variability between impressions originating from different fingers.

Nevertheless, we concur with Chacón, Kadane, Lauritzen and his colleagues, Mateu and Stern that directly modelling the distributions of selected variables measured on *minutiae* configurations will remove the issues that are related to the selection and use of the weighting function and will address some of the concerns that are expressed by Kadane and Stern.

One of us (Cedric Neumann) is currently working on a new model, which uses shape analysis for the selection and measurement of variables on *minutiae* configurations (this was also proposed by Hotz and Munk) (Neumann, 2010).

We see that the new model will calculate likelihood ratios in three stages. We start with a mark from the crime scene and ask the examiner to mark up those *minutiae* that are of sufficient quality for comparison purposes. Next, the examiner is presented with a control print: in casework this might arise from a designated suspect, or from the result of searching the mark against a database of prints. During the first stage, the examiner searches the control print to see whether there is a set of k *minutiae* that is sufficiently close to that selected from the mark (in the examiner's assessment) to justify a numerical assessment being carried out; the same process is carried out using an automatic fingerprint identification system algorithm to select the closest k *minutiae* configuration on each of the prints in the reference data set. During the second stage, the variability arising from distortion effects, or user influence, of the k-configuration selected on the control print is modelled and a probability density function is assigned to variables that are carefully selected to describe its shape. Similarly, a probability density function is assigned to model the variability of the shapes of the selected reference k-configurations. During the final stage, the probability density of the variables describing the shape of the mark configuration can be assigned for both the numerator and the denominator of the likelihood ratio.

To some extent, this new model will still rely on the selection of the k-configuration on each reference print that exhibits the most similarities with the mark configuration. Indeed, modelling the distribution of all possible k-configurations in a reference data set can only lead to a gross underestimation of the denominator of the model. Gastwirth and Pan, and Llewelyn noted that this selection is complex and potentially computer intensive. Vendors of automatic fingerprint identification systems have considered this problem for several decades and nowadays provide highly efficient algorithms. We plan to keep relying on this technology for this purpose.

In parallel, we are also working towards addressing the point that was made by Silverman, and by Giraldo and colleagues, by including more fingerprint features in our model. First, we realize that our model considers what we have called 'finger propositions' (Neumann *et al*., 2011a):

H_p : the mark was made by the finger that made the print;
H_d : the mark was made by some unknown finger.

However, when determining the source of a mark, examiners are considering 'person propositions':

H_p : the mark was made by the person who provided the set of control prints;
H_d : the mark was made by some unknown person.

In Neumann *et al.* (2011a), we propose an extension of the model to account for all 10 fingers of a suspected donor. Similarly, we have also proposed an extension of the model to account for the general pattern of the ridge flow that is observed on the mark (Neumann *et al.*, 2011b).

The addition of variables describing more minute details of finger impressions, such as shape and location of pores, or shape and width of ridge edges, is also being considered. Nevertheless, these elements of fingerprint patterns do not seem very reproducible, or carry much weight in addressing the issue of the source of a mark (Anthonioz *et al.*, 2008, 2011). In a majority of cases, these details are only present on marks that are of such quality that the abundance of more robust features makes them redundant. On lower quality marks, these features are usually used in practice to supplement the observation of few *minutiae* when forming one of the traditional categorical conclusions of fingerprint examination: identity of source or exclusion. The results that were presented in this paper show that, in those cases, considerable weight of evidence can already be obtained by only using *minutiae* features; in other words, weakly reproducible and discriminative features may not add much to the weight of evidence that is carried by low quality marks, and certainly not enough to form a categorical opinion of identity.

Concerns related to the validation of the model and its related databases have been brought up by several discussants. We agree with Balding that the questions of the size and representativeness of the reference database that is used in the model and of the influence of ethnicity, gender and relatedness on friction ridge features need to be addressed. As noted by Silverman, researchers need to access large data sets to strengthen the scientific foundations of forensic science. Because of the very specific nature of the field, most of these collections are government owned and contain personal identifier information. Access to these data sets remains an issue.

Balding, Fieller, Robertson and his colleagues, Stoney, Saks and Votruba, and Welling mention the need for calibrating the statistic resulting from the model. The discussants made valid points and this is certainly an area of growing interest (Morrison, 2011). Ultimately we come up against the notion of utility that was described by Biedermann *et al.* (2008).

Cole, Fienberg, Gastwirth and Pan, Hotz and Munk, Risinger, Saks and Votruba, and Thompson have considered the aspect of the deployment of the model. We agree with these discussants that the model itself is not a panacea for all criticisms and challenges made with regard to the examination of fingerprint evidence. It is clear that such a model can only be deployed together with proper standard operating procedures for the selection and labelling of fingerprint features to avoid bias and to prevent the model from being overly sensitive to human fingerprint examiners. Although in theory we like the idea of a completely automated process as suggested by Fienberg, and by Saks and Votruba, Fienberg himself describes the main challenge to this approach; the technology that is required to detect and select fingerprint features on partial and degraded marks is currently outperformed by human cognition capabilities. At the moment, we see that the labelling of features on mark and control prints needs to be performed by human examiners.

More generally, and as highlighted by Evett and Williams more than 20 years ago (Evett and Williams, 1996), the criticisms and challenges that have been raised against the forensic community can only be successfully answered by a global and coherent approach, which will include training, competency testing and accreditation of laboratories, in addition to technological development. On a related point, we thank Lauritzen and his colleagues for suggesting an approach to modelling confirmation bias.

As noted by Butt and Ramotoswki, we see our model as a tool that can support fingerprint examiners when forming conclusions. And together with Cole, Kaye, Saks and Votruba, and Risinger, we see such models as the means for progressing from the traditional paradigm of reporting the results of fingerprint examinations as categorical opinions. The use of more qualified conclusions will not only strengthen the scientific foundations of the examination of fingerprint evidence, but may also, as noted by Risinger, have a positive effect on the volume of fingerprint evidence that is currently reported. A recent field study (Neumann *et al.*, 2011c), performed with the model presented in this paper, has found that approximately 30% of the comparisons currently deemed inconclusive have the potential for providing quantitative assessments of evidential weight. The cost of the implementation of such a model in practice mentioned by Saks and Votruba would need to be weighed against the cost that is currently incurred by the community as a result of processing and comparing fingerprint evidence without reporting potentially useful observations.

Finally, Aitken, Silverman and Kaye raise the question of how an assessment of weight of evidence derived from a case analysis based on methods such as those described in our paper should be presented at court. This is, of course, a crucially important issue; particularly so given that judgements in the courts have, on several occasions, betrayed a poor understanding on the part of the judiciary of the principles of forensic inference. The R. *versus* T. judgement to which Aitken refers is only one of several. We applaud the

initiative that Aitken is leading, on behalf of the Royal Statistical Society, to improve the understanding of the legal profession as a whole of the logic of reasoning in the face of uncertainty within the forensic *milieu*.

Aitken points out that a numerical value alone will present a false impression of exactitude, yet a range of values will, no doubt, bring another set of problems. In the DNA field, the judgement in Doheny and Adams (R. *versus* Doheny and Adams, 1997, 1 Cr. App. R. 369) is interpretead (rightly or wrongly—the wording of the judgement is lamentably imprecise) to mean that, if the DNA expert can calculate a likelihood ratio, then it is the value alone that may be presented to the jury: the expert is not allowed to provide any verbal elaboration on the potential effect of that value within the context of the case. Yet we agree with Aitken when he says that the likelihood ratio in a fingerprint case should be tempered by the experience of the examiners. After all, the likelihood ratio depends crucially on a complex set of judgements that the examiner will have made. This relates to a point that we have made in the paper: it is a misconception, trivial situations apart, to believe that there is a 'correct' value of the likelihood ratio. In any area of forensic inference likelihood ratio is inevitably a function of the background data, the case observations, judgement by the operator and a set of modelling assumptions.

Ultimately, the question of presentation at court has multiple facets: we need to convey in the most efficient, yet fair, manner the weight of the evidence; we need to explain that the weight of evidence that is presented is the examiner's best attempt at providing a fair and reasonable assessment; and we need to be able to present all supporting assumptions and data used to assign the weight of evidence.

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